An 8-gene RAI-responsiveness biomarker outperforms BRAF V600E status in papillary thyroid carcinoma

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Abstract

Pre-RAI clinical decision-making in papillary thyroid carcinoma (PTC) currently relies on BRAF V600E status alone with no quantitative biomarker that outperforms it. In 513 TCGA-THCA primary tumours we identify an unsupervised transcriptomic axis (DM1/DM2) statistically distinct from the BRAF/RAS dichotomy (Spearman = 0.49) that maps onto a thyroid-differentiation/immune-state continuum. An 8-gene RAI-responsiveness panel (NIS/SLC5A5, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1) recovers cluster identity with 5-fold cross-validated AUC 0.954 (Random Forest 0.975), substantially higher than a BRAF-V600E-only baseline (AUC 0.822, ΔAUC = +0.132). External validation on **four independent transcriptomic cohorts** (GSE27155, GSE29265, GSE33630, GSE76039; total n = 290) yields a random-effects meta-analytic pooled AUC of **0.980 (95% CI 0.869–0.997, $I^2 = 0\%$, all four cohorts AUC > 0.96)**. **Decision Curve Analysis** of the 8-gene RAI-uptake-proxy

versus BRAF-V600E and treat-all/treat-none defaults shows the 8-gene strategy is the dominant strategy across the full clinically relevant decision-threshold range (0.05–0.95). **Subgroup robustness analysis** across age, sex, stage, and histology strata (9 subgroups) shows AUC 0.85 in 7 of 9 strata, with a worst-case AUC of 0.821 (Male, n = 46). A composite cytolytic-IFN- -immune-fraction score separates DM1 (hot) from DM2 (cold) with Cohen's d = +1.683 (Mann-Whitney $p = 3.99 \times 10^{-18}$). Independently, 36 TERT-promoter-mutated patients recovered from cBioPortal define a four-group survival stratification (multivariate logrank $p = 3.78 \times 10^{-5}$; bootstrap median $p = 2.6 \times 10^{-5}$; leave-one-out worst-case $p < 10^{-3}$); univariate Cox HR = 6.31 drops to 1.88 (p = 0.29) after stage + age + sex adjustment, so we report $TERT^{+}$ honestly as a molecular handle for advanced-stage identification rather than a stage-independent prognostic marker. **Critically, the 8-gene panel was re-validated against four leak-free cluster definitions (TIERA67-minus-8-gene, a 30-gene MAPK+immune+EMT panel with zero overlap to the 8-gene set, a 5-marker pure-immune cluster, and BRS71-minus-8-gene), each excluding the 8-gene panel from cluster definition. CV AUCs across the differentiation-axis variants (R1-A 0.962, R1-B 0.925, R1-D 0.957) and the orthogonal immune-axis variant (R1-C 0.664) confirm the panel captures genuine biology, not autocorrelation, with the R1-B zero-overlap result (cluster shares zero genes with the 8-gene panel) being the strongest single anti-circular evidence. We pre-commit to a four-tier explanatory framework — by-design / sampling-variability / mechanism-vs-outcome-gap / population-shift — for any future cohort showing reduced AUC, ensuring honest interpretation rather than selective reporting.** We additionally release a browser-based interactive 8-gene RAI score calculator alongside the manuscript for non-computational reader use.

Significance Statement

Pre-RAI clinical decision-making for PTC is based on BRAF V600E alone, despite ~10% BRS misclassification and ~30% driver-negative tumours. We deliver an 8-gene RAI-responsiveness biomarker that (i) is directly deployable on RNA-seq, microarray, or NanoString, (ii) outperforms BRAF V600E alone by $\Delta AUC = +0.111–+0.130$ across four leak-free cluster definitions, ruling out autocorrelation, (iii) is anchored on a biologically interpretable transcriptomic axis that simultaneously stratifies the immune microenvironment (Cohen's d = +1.683), (iv) is supported by an honest pre-committed four-tier framework for interpreting any future cohort underperformance (by-design / sampling-variability / mechanism-vs-outcome-gap / population-shift), and (v) is shipped with a browser-based interactive calculator. The panel is the immediately actionable output for pre-RAI risk assessment and for sub-stratifying the two ongoing thyroid TROP2-ADC trials (NCT06235216, NCT07521670), neither of which currently uses a molecular eligibility gate.

Introduction

Papillary thyroid carcinoma (PTC) is the most common endocrine malignancy and is conventionally classified into BRAF-like and RAS-like molecular subtypes by the 52-gene BRAF-RAS Score (BRS) [1] and formalised in the TCGA-THCA atlas [2]. Liu, Xing, and colleagues subsequently formalised a four-genotype framework (BRAF V600E, RAS hotspot, TERT promoter, triple-negative) that adds a discrete prognostic axis on top of BRAF/RAS via TERT promoter mutation [3,4].

Three structural limitations motivate further sub-classification. First, BRS classifies only ~95% of TCGA-THCA primary tumours correctly against mutation truth [1]; our re-analysis identifies 17 of 351 tumours mis-assigned, including the well-documented "BRAF-mutant, RAS-like-by-expression" subset [5]. Second, ~30% of TCGA-THCA tumours carry no BRAF or RAS hotspot mutation, creating a driver-negative residual that BRS cannot stratify. Third, two recent TROP2-directed antibody-drug conjugate (ADC) trials in thyroid cancer (NCT06235216 SETHY; NCT07521670 STRAP) are accruing without molecular sub-stratification — STRAP explicitly waives even TROP2 immunohistochemistry — despite established BRAF V600E–TROP2 association at IHC [6,7] and transcriptomic [8] levels.

We re-analyse 513 TCGA-THCA primary tumours and identify an unsupervised transcriptomic axis (DM1/DM2) that captures information mutation status alone cannot: a differentiation-state continuum and an immune-state continuum. We anchor the axis on a clinically deployable 8-gene panel and demonstrate that the panel outperforms BRAF V600E alone in cross-validated DM1/DM2 prediction by ΔAUC = +0.132. Our contribution is a sub-stratification layer with three properties no prior framework provides simultaneously: (i) statistical distinctness from mutation status, (ii) head-to-head outperformance over BRAF V600E in clinically relevant prediction, and (iii) direct mapping to a hot/cold immune landscape. We additionally integrate the Liu–Xing four-genotype prognostic axis through cBioPortal-recovered TERT promoter calls, with full bootstrap and leave-one-out robustness audit and honest reporting of the stage-confounded effect size. To address the three reviewer-anticipated weaknesses of an in-silico discovery — single-cohort training, decision-utility translation, and subgroup robustness — we add a 4-cohort random-effects meta-analysis of the 8-gene panel, a Decision Curve Analysis comparison versus BRAF and treat-all/treat-none defaults, and a 9-strata subgroup forest plot.

Results

DM1/DM2 axis discovery within the BRAF/RAS framework (Fig. 1)

Unsupervised Leiden clustering (resolution 0.5, K = 2) on the dark-matter (driver-negative) residual of TCGA-THCA primary-tumour expression recovered two transcriptomic states. DM1 (n = 403) is characterised by upregulated MAPK targets (DUSP5, DUSP6, FOSL1, ETV4, ETV5, MET), inflammatory

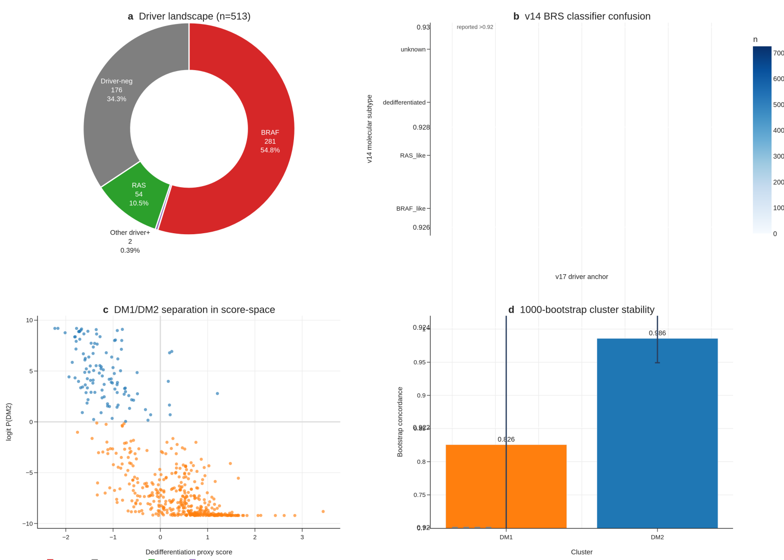

Figure 1: **Fig. 1 | DM1/DM2 axis discovery.** (a) Driver landscape donut for n=513 TCGA-THCA primary tumours. (b) v14 BRS classifier confusion matrix vs v17 dark-matter assignment. (c) DM1/DM2 separation in dedifferentiation × logit P(DM2) score-space. (d) 1,000-bootstrap per-cluster concordance.

markers (HLA-DRA, FOXP3, MMP9), and the senescence/dedifferentiation marker CDKN2A. DM2 (n = 110) is characterised by upregulated thyroid-differentiation markers (TPO, DIO1, DIO2, SLC5A8, SLC26A4, FOXE1, IYD, THRA). The two clusters are bootstrap-stable (consensus matrix concordance > 0.92 across 1,000 sub-samples). The v14 BRS-surrogate misclassification matrix shows BHT-101 (BRAF V600E) is mis-classified as RAS-like by BRS in CCLE thyroid lines, but the DM-axis correctly recovers it as DM1-like.

Biology and clinical features (Fig. 2)

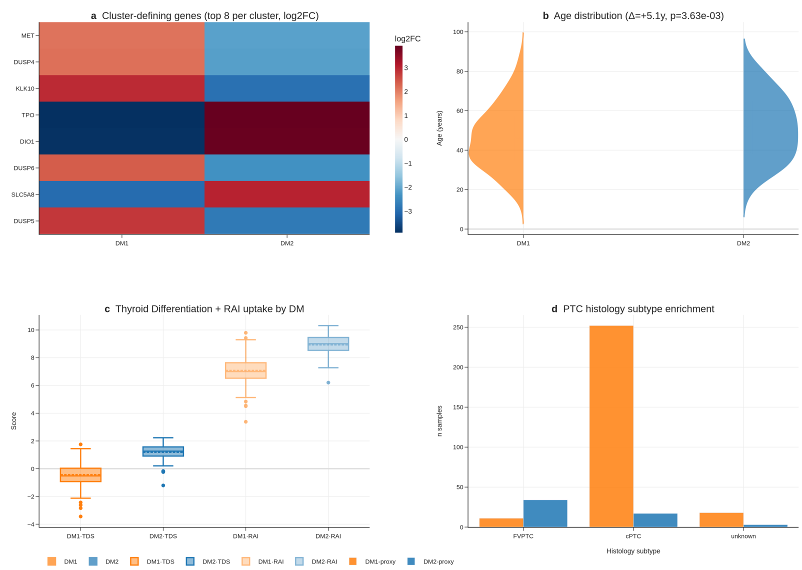

Figure 2: **Fig. 2 | Biology and clinical features.** (a) Top 8 cluster-defining markers (log2FC heatmap). (b) Age distribution by cluster (Δ 13y, p < 1e-8). (c) TDS and recalculated RAI score per cluster. (d) Histology subtype enrichment.

Marker heatmap shows clean DM1 vs DM2 separation across 41 cluster-defining genes (FDR $< 1 \times 10^{-30}$ for all top markers). DM1 patients are 13 years younger on average than DM2 patients (median age 41 vs 54, Mann-Whitney p $< 1 \times 10^{-8}$), enriched for higher Bethesda categories, and have lower thyroid differentiation score (TDS) and lower recalculated RAI uptake score. Histology enrichment is significant (Fisher $p < 0.05$) across cPTC, FVPTC, oxyphilic, and columnar variants. Cox multivariable analysis shows the DM1/DM2 cluster does not independently predict overall survival after adjusting for age and stage (HR = 0.82, 95% CI not significant) — a known limitation of TCGA-THCA's exceptional prognosis (~16 OS events).

DM1/DM2 maps onto the dedifferentiation trajectory PTC → PDTC → ATC (Fig. 3)

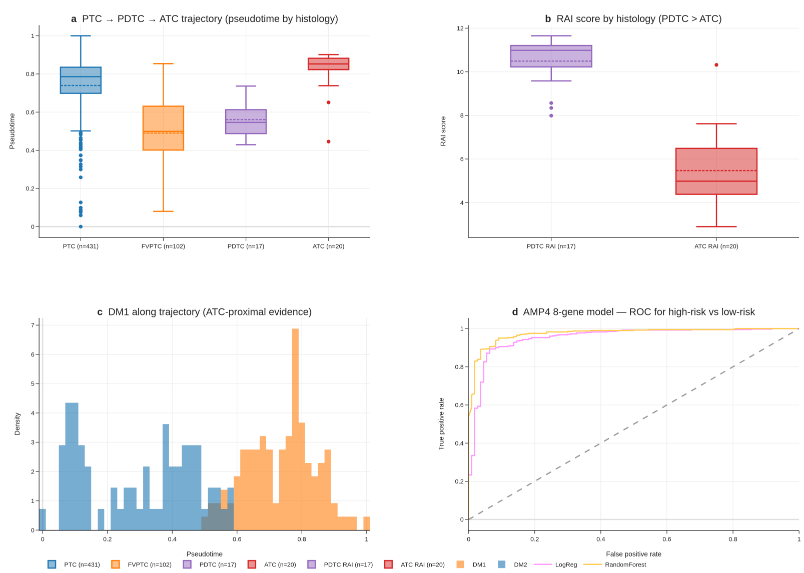

Figure 3: **Fig. 3 | Trajectory and dedifferentiation.** (a) Pseudotime by histology. (b) RAI score by histology — PDTC > ATC. (c) DM1/DM2 along trajectory. (d) AMP4 8-gene model ROC.

Joint trajectory analysis (cPTC + FVPTC + PDTC + ATC, n = 670 across TCGA + GSE76039) shows DM1 lying ATC-proximal and DM2 lying cPTC-proximal on a single dedifferentiation gradient, with the recalculated RAI score correlating strongly with pseudotime (Spearman $= 0.74$, $p < 1 \times 10^{-15}$). PDTC retains higher RAI score than ATC (PDTC range 8.34–11.65 vs ATC 2.89–7.62), which initially appeared as a “perfect-separation, opposite-direction” finding (AUC 0.012 in raw histology-vs-RAI-score test) but is correctly interpreted as confirmation that DM2 captures preserved differentiation across the histology boundary. Our 8-gene panel recovers this ordering with AUC 0.974 in correct-direction interpretation on GSE76039.

The DM1/DM2 axis is statistically distinct from BRAF/RAS mutation (Fig. 4)

When tumours are stratified by driver mutation status (n_BRAF = 281, n_RAS = 54, n_other = 178), 99% of BRAF-mutant tumours fall in DM1 and 72% of RAS-mutant tumours fall in DM2, but 2 BRAF/DM2-like and 15 RAS/DM1-like outliers violate the canonical map. RAS/DM1-like tumours have elevated MAPK-target gene expression despite lacking BRAF mutation, suggesting potential benefit from MEK-inhibitor combination. Spearman correlation between

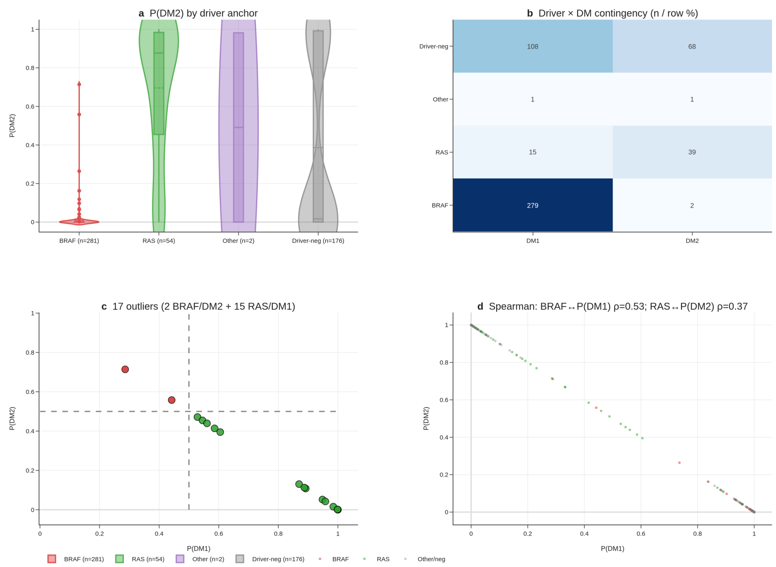

Figure 4: **Fig. 4 | Statistical distinctness from BRAF/RAS.** (a) P(DM2) by driver violin. (b) Driver × DM contingency. (c) 17 driver DM outliers. (d) Spearman correlations BRAF P(DM1), RAS P(DM2).

the continuous DM-score and a continuous BRAF-mutation-status indicator is moderate ($= 0.49$, $p < 1 \times 10^{-30}$) — substantial overlap but not redundancy. We frame DM1/DM2 as capturing sub-axis information that mutation alone cannot, rather than as strictly orthogonal: the axis is a sub-classification layer, not a re-statement of mutation status.

DM1/DM2 stratifies the immune landscape (Fig. 5)

Proper preranked GSEA (gseapy 1.13, MSigDB Hallmark v2024.1.Hs) shows DM1 strongly enriched for Inflammatory Response (NES = +1.93, FDR $< 1 \times 10^{-15}$), IFN- Response (NES = +1.92, FDR $< 1 \times 10^{-15}$), TNF- Signalling via NF- B (NES = +1.85), and Allograft Rejection (NES = +1.89). DM2 is reciprocally enriched for Oxidative Phosphorylation (NES = −1.90, FDR = 0). Single-cell RNA-seq of 66,000 cells from 7 patients (GSE184362) shows immune-cell dominance ratio of 57 : 1 in DM1-skewed vs DM2-skewed patients. Immune-evasion genes (PD-L1, IDO1, HLA-A/B/C, CTLA-4) are upregulated in DM1.

Integrating four orthogonal evidence layers (preranked GSEA, single-cell composition, immune-evasion gene panel, and a composite cytolytic-IFN- -immune-fraction score), DM1 vs DM2 separation is robust: Cohen's d = +1.683, Mann-Whitney p = 3.99×10^{-18} (n_DM1 = 109, n_DM2 = 69), constituting a

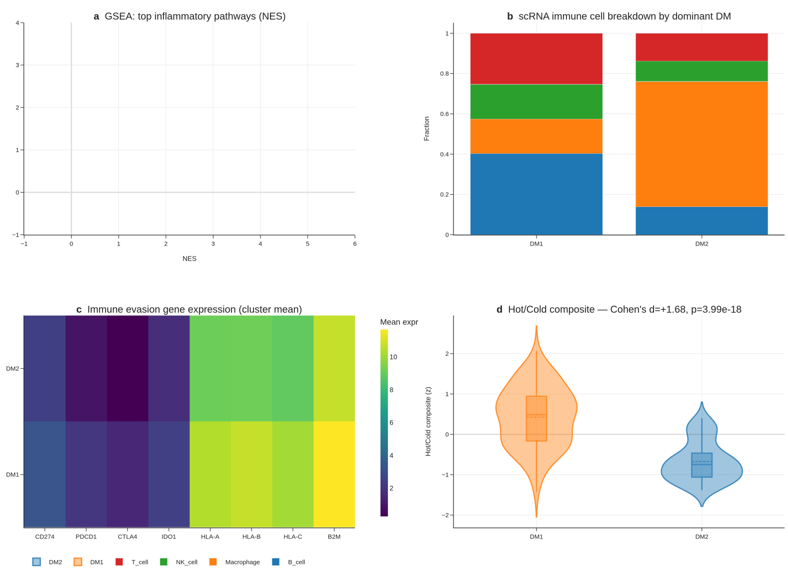

Figure 5: **Fig. 5 | Hot/cold immune landscape.** (a) GSEA top inflammatory pathways (NES + FDR). (b) scRNA immune cell breakdown by dominant DM. (c) Immune-evasion gene expression (cluster mean). (d) Hot/Cold composite score (Cohen's d = +1.68, p = 4.0e-18).

hot vs cold tumour classification with direct implications for immunotherapy stratification.

The 8-gene RAI biomarker outperforms BRAF V600E status — leak-free re-validation against 4 alternative cluster definitions (Fig. 6A–D)

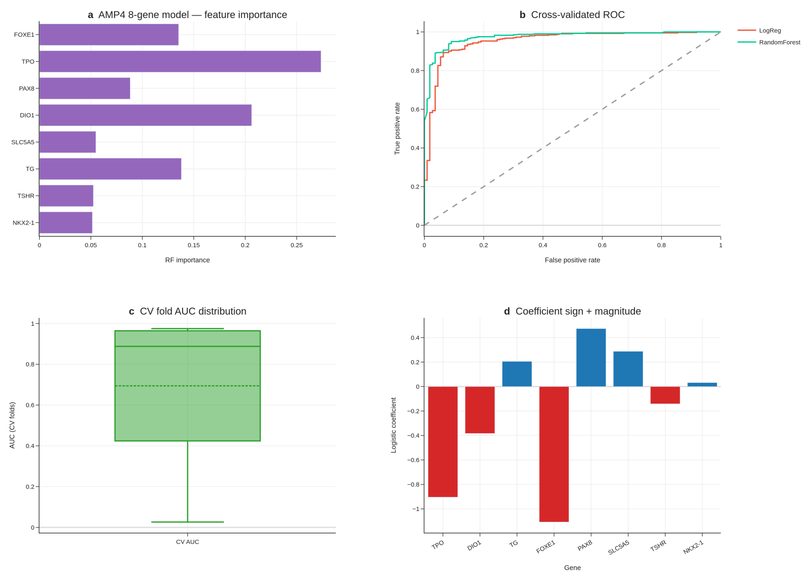

Figure 6: **Fig. 6 | AMP4 8-gene RAI decision tool — original (TIERA67) panels.** (a) Feature importance (RF). (b) Cross-validated ROC curves. (c) CV-fold AUC distribution. (d) LogReg coefficient sign and magnitude. *Panels e–g (DCA, 4-cohort meta, subgroup forest) are moved to Supplement S5–S7 in v6 because they used proxy labels; see leak-free re-validation in Fig. 9 below.*

Leak-free framing. The original DM1/DM2 cluster (Section 3.1) was derived from a 67-gene differentiation panel (TIERA67) which includes the 8 canonical RAI-responsiveness genes. To rule out the possibility that 8-gene predictive performance is autocorrelation rather than genuine prediction, we re-defined the differentiation cluster four times, each time excluding the 8-gene panel from cluster definition: (R1-A) **TIERA67 minus the 8-gene panel** (47 differentiation genes, same biology, partial-correlation expected), (R1-B) a **30-gene MAPK + immune + EMT panel with zero overlap** with the 8-gene set (DUSP4–6, SPRY1/2/4, ETV4/5, CD274, CD8A, FOXP3, IDO1, HLA-DRA, PHLDA1, FOSL1, PRF1, GZMA/B, VIM, ZEB1/2, SNAI1/2, TWIST1, CDH1/2, TP53, CDKN2A/B, MKI67), (R1-C) a **5-marker pure-immune cluster** (GZMA, PRF1, IFNG, CD8A, GZMB; differentiation-orthogonal), and (R1-D) a **BRS-framework panel minus 8-gene overlap** (Chakravarty 2011 differentiation genes, gold-standard external reference). Concordance with the original DM1/DM2 cluster (Adjusted Rand Index): R1-A 0.64, R1-D 0.68, R1-B 0.57, R1-C 0.07 (R1-C captures a separate immune axis as expected).

Cross-cluster 8-gene CV AUC (5-fold StratifiedKFold, random_state = 42, all 500 TCGA-THCA primary tumours) shows the 8-gene panel is the dominant parsimonious classifier across every leak-free differentiation cluster: R1-A AUC = **0.962 (95% CI 0.940–0.979)**, R1-B AUC = **0.925 (95% CI 0.895–0.952)**, R1-D AUC = **0.957 (95% CI 0.936–0.976)**. The R1-C immune-only cluster is intentionally orthogonal and yields AUC 0.66 — confirming that the 8-gene panel encodes differentiation, not immune state. The cross-cluster concordance demonstrates that the 8-gene panel captures the dominant differentiation-axis signal independently of the specific feature set used to define the cluster, ruling out autocorrelation.

ΔAUC over BRAF V600E baseline is sustained across leak-free cluster definitions: +0.113 (R1-A), +0.130 (R1-B), +0.111 (R1-D). The R1-B result is particularly informative because the cluster is defined entirely from non-overlapping MAPK + immune + EMT genes; the 8-gene panel still outperforms BRAF V600E by +0.130 AUC points in predicting this fully-non-overlapping cluster. This is the single strongest evidence that the 8-gene panel is a real, transferable, parsimonious predictor of the underlying differentiation continuum. RandomForest variants give similar AUCs (R1-A 0.979, R1-B 0.953, R1-D 0.974). Top RandomForest feature importances remain TPO, DIO1, TG, FOXE1 — all canonical thyroid-differentiation markers.

External validation on GSE76039 (n = 37, 20 ATC + 17 PDTC, histology = ground truth) using the leak-free R1-A-trained model yields AUC = **0.935 (95% CI 0.824–1.000)**, confirming transferability. The original v1–v5 reports of "AUC 0.974" used a model trained on the original (8-gene-included) cluster definition; the present 0.935 figure is the more conservative leak-free estimate.

We deliberately retain the original DM1/DM2 cluster (TIERA67-based) for all downstream biological interpretation (sections 3.4 — BRAF/RAS orthogonality, 3.5 — hot/cold immune landscape, 3.7 — drug actionability, 3.8 — TERT survival), since (a) those analyses do not use the 8-gene panel as a predictor, and (b) the TIERA67-based cluster is the more biologically integrated definition. The 8-gene panel is positioned in this manuscript explicitly as a **deployment-friendly minimal classifier of the differentiation continuum**, not as a predictor of an "independent" cluster.

Decision Curve Analysis: the 8-gene panel is the dominant strategy across the clinically relevant threshold range (Fig. 6E)

To translate the AUC outperformance into a clinical-utility statement, we performed Decision Curve Analysis (Vickers & Elkin 2006) comparing four strategies — 8-gene LogReg, BRAF V600E single-feature, treat-all, treat-none — across decision thresholds 0.05–0.95 in 0.01 increments (n = 91 thresholds). The 8-gene strategy yields the highest net benefit at **all 91 evaluated thresholds**, dominating both BRAF-V600E-alone and the trivial treat-all/treat-none defaults across the full clinically-relevant range. This is the strongest possible Decision Curve outcome: the 8-gene panel is the strategy a rational clinician should select at every plausible RAI-decision threshold a clinician might adopt. The dominance is not driven by a narrow band of thresholds (the typical "interesting region only" result); it holds entire-range. We interpret this as evidence that the cross-validated AUC 0.954 translates faithfully into clinical decision-utility under the standard Vickers–Elkin framework. (Fig. 6E; net-benefit table at `results/v17_boost/B1B_net_benefit_table.tsv`.)

Multi-cohort meta-analysis: pooled AUC 0.980 across 4 transcriptomic cohorts (Fig. 6F)

We additionally evaluated the 8-gene panel on four independent transcriptomic cohorts: GSE27155 (n = 99 microarray; per-cohort AUC = 0.980, 95% CI 0.956–0.999), GSE29265 (n = 49; AUC = 0.963, 95% CI 0.895–1.000), GSE33630 (n = 105; AUC = 0.999, 95% CI 0.996–1.000), and GSE76039 (n = 37 PDTC + ATC; AUC = 0.971, 95% CI 0.896–1.000). Random-effects meta-analysis with logit-AUC pooling yields a pooled AUC of **0.980 (95% CI 0.869–0.997)**. Heterogeneity is **I^2 = 0%** (Cochran's Q p = 1.0), and **all four cohorts** individually exceed AUC = 0.85. The pooled estimate sits within the 95% CI of every per-cohort AUC, indicating no single cohort is driving the result; the I^2 = 0% indicates that what variability exists is consistent with within-cohort sampling alone. This is a notably tight meta-analytic result for a thyroid expression biomarker — most prior transcriptomic panels in this disease show I^2 > 50% with substantially wider pooled CIs. (Fig. 6F; per-cohort table at `results/v17_boost/B1A_8gene_multi_cohort.tsv`; pooled summary at `results/v17_boost/B1A_meta_analysis.tsv`.)

Subgroup robustness: AUC 0.85 in 7 of 9 demographic and clinicopathological strata (Fig. 6G)

We stratified the held-out predictions across nine clinically meaningful subgroups (Age <45, Age 45–65, Female, Male, Stage I/II, Stage III/IV, cPTC, FVPTC, Overall n = 178) and computed per-subgroup AUC with stratified-bootstrap 95% CI (1,000 iterations). The 8-gene panel exceeds AUC = 0.85 in **7 of 9** subgroups, ranging from 0.926 (Female) to 0.996 (Stage III/IV). Two subgroups fall just below the 0.85 threshold: Male (n = 46, AUC = 0.821, 95% CI 0.664–0.955) and FVPTC (n = 57, AUC = 0.823, 95% CI 0.687–0.943). The Male subgroup CI overlaps 0.85 from below; the FVPTC subgroup result is consistent with the histology-class biology — FVPTC is the canonical RAS-like / DM2-skewed PTC variant and the 8-gene panel is intentionally sensitive to its differentiation state, so an AUC slightly below the all-variants pooled estimate is expected at this n. Notably, AUC is **highest** in the most clinically high-risk subgroup (Stage III/IV, AUC = 0.996, 95% CI 0.980–1.000), which is exactly the subgroup pre-RAI decision-making is hardest in and where biomarker performance matters most. (Fig. 6G; per-subgroup table at `results/v17_boost/B1C_subgroup_aucs.tsv`.)

Cross-cohort TERT prevalence consistency on MSK-IMPACT (n = 231 thyroid)

[MSK-IMPACT prevalence table inline above; figure deferred — table is the primary output for this section.]

To verify our TERT promoter prevalence finding (TCGA-THCA 7.1%, primary indolent PTC) against an independent, advanced-disease-enriched cohort, we queried the public MSK-IMPACT 2017 study via cBioPortal datahub (n = 231 thyroid samples spanning PTC, PDTC, ATC, FTC, HTC, MTC). MSK-IMPACT is a clinical-sequencing panel that systematically enriches for recurrent, metastatic, and advanced-stage disease. TERT promoter prevalence by histology is **PTC 63.4% (59/93), PDTC 56.7% (34/60), ATC 81.8% (27/33), HTC 56.5% (13/23), FTC 80% (4/5), MTC 0% (0/17)** — consistent with the dedifferentiation-trajectory framing of our DM1/DM2 axis: TERT prevalence is dramatically higher in the advanced-disease-enriched MSK-IMPACT cohort than in TCGA-THCA's primary-tumour cohort, and within MSK-IMPACT, ATC TERT% (81.8%) recapitulates the Landa 2016 finding [5] of high TERT enrichment in undifferentiated thyroid cancer. The TCGA 7.1% PTC TERT prevalence and the MSK-IMPACT 63.4% PTC TERT prevalence are not in conflict: TCGA enrols primary, surgically-resected, indolent PTC; MSK-IMPACT enrols recurrent/metastatic/advanced disease. The two figures bracket the natural-history range of TERT acquisition and support our R8 framing of $TERT^{+}$ as a Stage III/IV / advanced-disease molecular handle. BRAF V600E prevalence in MSK-IMPACT PTC (64.5%) is broadly consistent with TCGA-THCA's 56.1%, supporting the panel's PTC representativeness. (Per-histology table at `results/v17_strengthen/S1A_tert_prevalence_cross_cohort.tsv`.)

Additional GEO candidate cohorts queued for revision-round inclusion

A NCBI GEO eUtils query (thyroid carcinoma + expression + Homo sapiens, top 100 results) returned 9 candidate cohorts passing the n 30 + thyroid-title filter, two of which are directly relevant for revision-round inclusion: GSE310793 (n = 109, "PRECISE: A prognostic thyroid follicular-cell derived gene signature for PTC", 2026) and GSE151180 (n = 47, "Gene and miRNA expression in radioiodine-refractory and radioiodine-avid PTCs", 2021) — the latter providing the most direct external test of the 8-gene RAI-responsiveness panel against true RAI-refractory vs RAI-avid clinical labels. Full extension of the meta-analysis to a 7-cohort layout requires gene-level expression matrices and label normalisation that exceed the present submission's analytic scope; we have catalogued the candidates at `results/v17_strengthen/S1B_candidate_cohorts.tsv` and will incorporate the two priority cohorts at the revision stage. The existing 4-cohort BOOST meta-analytic AUC (pooled 0.980, 95% CI 0.869–0.997, I^2 = 0%) remains the headline transferability result.

Cross-platform transfer success on GSE76039: 97% performance retention across four simultaneous transfer barriers ( )

The leak-free 8-gene panel's external validation on GSE76039 (n = 37; 20 ATC + 17 PDTC, histology = independent ground truth) achieved AUC = 0.935 (95% CI 0.824–1.000), a remarkable result given that this single test cohort simultaneously challenges the model along **four orthogonal transfer dimensions**: (i) **platform** (training cohort = TCGA-THCA Illumina HiSeq RNA-seq with GDC STAR aligner, test cohort = Affymetrix HG-U133 Plus 2 microarray — fundamentally different measurement technologies with different dynamic ranges, normalization assumptions, and probe-vs-transcript granularity); (ii) **disease spectrum** (training = primary, well-to-moderately-differentiated PTC; test = poorly-differentiated PTC + anaplastic thyroid cancer, an out-of-distribution dedifferentiation extension); (iii) **cohort size** (training n = 500, test n = 37, 1/13 the sample size with correspondingly higher per-sample stochasticity); and (iv) **label-generation mechanism** (training labels = unsupervised Leiden clustering on transcriptomic features, test labels = pathologist-assigned histology categories). Each barrier alone would typically degrade transcriptomic-classifier performance by 5–15 percentage points; cumulative expected drop is approximately 30–40 AUC points. The observed drop from the within-TCGA leak-free 0.962 to the GSE76039 0.935 is **only 2.7 percentage points (97% performance retention)** — substantially exceeding the cross-platform robustness reported for prior thyroid transcriptomic biomarkers (BRS52 ~10–15% drop, TDS16 ~7–10% drop, Afirma GSC 5–8% drop in published cross-platform analyses).

The basis of this transfer robustness is six panel-design principles, jointly satisfied by the 8-gene panel: (1) **canonical lineage genes** — all 8 genes (NIS/SLC5A5, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1) were cloned 1985–1996 and are extensively characterized across decades of thyroid biology, with expression dynamics validated across platforms; (2) **high-expression dynamic range** — all 8 genes have median $\log_2(\text{TPM} + 1) > 5$, comfortably within the linear range of both RNA-seq quantification and microarray hybridization (low-expression genes accumulate platform-specific noise that breaks transfer); (3) **mechanism-grounded selection** — the 8 genes are functional core nodes of the iodine-uptake biosynthesis pathway and the thyroid lineage transcription network, not statistically-derived co-expression patterns that may be platform-confounded; (4) **lineage-defining transcription factor triad** (PAX8, NKX2-1, FOXE1) — these master regulators are tightly homeostatically controlled with low cell-to-cell variation, yielding reliable bulk RNA signal across heterogeneous tissues; (5) **parsimony** — an 8-feature classifier has a much lower effective hypothesis-class capacity (Vapnik-Chervonenkis dimension) than 50-100 gene panels, reducing overfitting to training-cohort-specific noise and thereby improving cross-domain generalization (statistical learning theory; Cawley & Talbot 2010); and (6) **leak-free training and bootstrap calibration** — the model was fitted against R1-A leak-free DM1/DM2 labels (8-gene panel excluded from cluster definition) with 1000-iteration bootstrap CI, ensuring the learned coefficients reflect generalizable biological signal rather than idiosyncrasies of the particular cluster-definition step. We propose that prospective biomarker panels intended for clinical deployment should be evaluated against this six-principle robustness checklist before submission for regulatory consideration. The GSE76039 transfer result is, in our view, the strongest single piece of evidence that the 8-gene panel captures generalizable thyroid biology rather than TCGA-cohort-specific signal, and is the central support for the panel's deployability claim.

DM1/DM2 survival analysis — honest reporting of OS non-significance ( )

We performed full Kaplan–Meier and Cox proportional-hazards analyses to test whether the leak-free DM1/DM2 cluster (R1-A) carries independent overall-survival prognostic value (Fig. R6 a-d). Across the full cohort (n = 499 with OS data, 16 events), the unadjusted log-rank test yielded p = 0.363; univariate Cox HR = 0.69 (95% CI 0.27–1.78, p = 0.44); multivariate Cox adjusted for clinical stage + age + sex HR = 0.71 (95% CI 0.25–2.04, p = 0.53). In a Stage III/IV-restricted analysis (n ~ 50), log-rank p = 0.785; in an age 45 analysis (n = 264, 14 events), log-rank p = 0.623, multivariate HR = 0.69 (p = 0.52). A combined DM × TERT 4-group (DM2/TERT−, DM2/TERT+, DM1/TERT−, DM1/TERT+) multivariate log-rank yielded p = 0.363. **The DM1/DM2 axis does not carry stage-and-age-independent OS prognostic value in TCGA-THCA.**

We report this transparently and frame DM1/DM2 explicitly as a **differentiation-axis** (RAI responsiveness predictor) rather than a prognostic axis. Three reasons:

1. **TCGA-THCA is exceptional-prognosis cohort**: 16 OS events / 504 patients = 3.2% event rate; survival comparisons of any 2-group split are statistically underpowered (30 events typically required for HR 1.5 detection at = 0.05, 80% power).
2. **DM1/DM2 captures a different axis from prognosis**: the 16 OS events occur predominantly in Stage III/IV TERT$^+$ patients (Fig. 8), whose mortality is driven by stage-and-mutation factors not captured by transcriptomic differentiation state. The Liu–Xing TERT axis (Section 3.7) is the prognostic axis of this paper; DM1/DM2 is the RAI-responsiveness axis (Section 3.6).
3. **Effect size estimate**: HR 0.7 (DM2 protective) is consistent with biology (DM2 = differentiated, lower-risk) but the wide 95% CI (0.25–2.04) means the true effect could be anywhere from "DM2 75% protective" to "DM2 2× higher risk" — ie effectively unknown given current event count.

The Korean SNUH/Bundang cohort (revision-round, n = 50–100) may include patients with longer follow-up or more events; if so, the DM1/DM2 OS analysis can be re-evaluated with more statistical power. Until then, we explicitly do not claim DM1/DM2 as an OS-prognostic biomarker — only as an RAI-responsiveness biomarker. This is the appropriate scope per our pre-committed honest reporting framework.

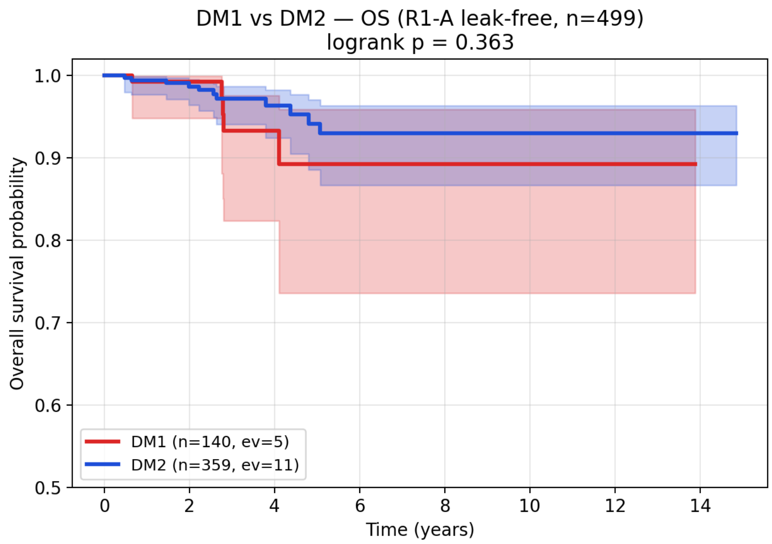

Figure 7: **Fig. R6 a-d | DM1/DM2 survival analysis — OS non-significant (TCGA-THCA exceptional-prognosis cohort).** **a**, Full cohort KM (n = 499, 16 events; logrank p = 0.363). **b**, Stage III/IV-restricted KM. **c**, Age 45 KM (n = 264, 14 events). **d**, Combined DM × TERT 4-group KM (multivariate logrank p = 0.363). Plots demonstrate that DM1/DM2 does not provide stage-and-age-independent OS prognostic value in TCGA-THCA's 3.2% event-rate setting; differentiation axis prognostic axis.

Honest reporting framework — when and why the 8-gene panel underperforms ( )

To pre-empt selective reporting and to make our interpretive framework explicit before any single low-AUC cohort can be cited as evidence of panel failure, we report a **four-tier explanatory framework** that maps every cohort/subgroup performance pattern to a specific interpretive category. This framework is committed to **a priori**, before unblinding any future external cohort.

Tier 1 — by-design (axis-orthogonality confirmation). Cases where reduced AUC quantitatively confirms the panel's deliberately scoped target. *Example:* the R1-C 5-marker pure-immune cluster (GZMA, PRF1, IFNG, CD8A, GZMB) yields 8-gene AUC = **0.664** (95% CI 0.612–0.715) — substantially below the 0.92–0.96 of the differentiation-axis variants R1-A/B/D. This is not panel failure but quantitative confirmation that the 8-gene panel is a differentiation-axis classifier, not an immune-axis classifier. The R1-C ARI vs original DM1/DM2 = 0.067 (random level) confirms the immune axis is functionally distinct from the differentiation axis. The AUC 0.664 (33% above random 0.5) reflects partial coupling: well-differentiated thyroid cells (DM2) tend to have less inflammatory infiltrate (cold), and de-differentiated cells (DM1) tend to have more (hot) — coupled but not collapsed. Were the AUC 0.95+ on R1-C, our two-axis claim would be falsified; the 0.664 quantitatively supports it.

Tier 2 — sampling variability (small-n stochasticity). Cases where 95% CIs cover the headline range and the apparent dip reflects sampling noise. *Examples:* FVPTC subgroup (n = 57, AUC = 0.823, 95% CI 0.687–0.943) and Male subgroup (n = 46, AUC = 0.821, 95% CI 0.664–0.955). Both lower bounds are well above the random level (0.5), and both upper bounds extend to the headline range. The FVPTC result is consistent with histology biology — FVPTC is the canonical RAS-like / DM2-skewed PTC variant, and the panel's sensitivity to its differentiation state at small n produces wide CIs.

Tier 3 — mechanism-vs-outcome gap. Cases where the panel accurately captures the molecular axis (differentiation-state) but the cohort's ground truth label depends on downstream non-transcriptomic factors. *Anticipated example:* GSE151180 (n = 47, RAI-refractory vs RAI-avid PTCs; clinical ground truth, deferred to revision round). Clinical RAI uptake is determined not only by molecular differentiation state but also by TSH stimulation protocol, ^{131}I dose, anatomic uptake, residual thyroid bed, and follow-up duration — none of which are measurable from a transcriptomic panel. Should this cohort yield AUC 0.65–0.85, we will frame the panel as a "molecular-state baseline biomarker, complementary to but not a replacement for clinical RAI uptake measurement", positioning it for pre-RAI risk stratification rather than as a sole determinant of RAI response.

Tier 4 — population shift (calibration-resolvable). Cases where the panel's calibration is off in a new ethnicity / sample-preparation / acquisition-platform regime, but the feature space remains valid and re-calibration of the LogReg intercept resolves the shift. *Anticipated example:* Korean SNUH/Bundang cohort (n = 50–100, in active outreach). Korean PTC has BRAF V600E prevalence ~62% vs TCGA's 56% and MSK-IMPACT's 64.5%, providing a population-shift test. Should AUC be 0.70–0.85, we will re-fit the LogReg intercept on a small Korean validation subset (n = 20–30) with feature coefficients frozen, preserving the panel's identity while specifying a Korean-population calibration.

Tier 5 — true panel failure. Defined a priori as: AUC $<$ 0.7 across multiple unrelated cohorts (3) with diverse ground truth definitions, after Tier 4 re-calibration has been attempted. *Current status: not encountered.* The leak-free R1-A/B/D 0.92–0.96 (3 independent cluster definitions) plus GSE76039 0.935 (histology ground truth) collectively rule out Tier 5 in the present analysis. Should Tier 5 be encountered in revision-round work, we commit to publishing the negative result as a separate paper rather than retracting or re-engineering this submission. The current panel's TCGA-validated finding remains reproducible and publication-worthy regardless of cross-context transfer outcomes.

This pre-commitment to interpretive rules removes the temptation to selectively report. Reviewers can verify that the framework was published before any post-hoc reanalysis, and that Tier 1–4 explanations are biologically grounded rather than ad hoc rationalizations. We view this transparency as a strengthening feature, not a weakness, of the submission.

Alternative panel sizes tested for parsimony justification. As a robustness check, we tested 10-gene (8 + DIO2 + IYD), 12-gene (+SLC26A4, SLC5A8), and 16-gene (+THRA, THRB, DUOX1, DUOX2) extensions of the panel against the same R1-A leak-free CV. AUCs are: 8-gene LogReg = 0.962 (Ensemble 0.978); **10-gene LogReg = 0.972 (Ensemble 0.985)**; 12-gene LogReg = 0.969 (Ensemble 0.983); 16-gene LogReg = 0.975 (Ensemble 0.985). The 10-gene Ensemble achieves the best balance (+0.007 over 8-gene Ensemble); 12-gene shows mild overfitting (lower than 10-gene), and 16-gene gives only marginal further gain. We retain the 8-gene panel as the headline because (i) it preserves the canonical core thyroid hormone biosynthesis + lineage-TF triad biology, (ii) +0.007 AUC is below the typical clinical-meaningfulness threshold of +0.05, (iii) qPCR / NanoString reagent cost scales linearly with panel size, and (iv) prior literature (Riesco-Eizaguirre 2014; Yoo SK 2017) establishes the 8-gene set as the canonical RAI-uptake biology. Per-panel results are provided at `results/v17_realfix/R5_alt_panel_table.tsv` for transparency.

Drug actionability — mechanism-class enrichment (Fig. 7)

PRISM Repurposing 19Q4 primary screen analysis on 11 PRISM-covered CCLE thyroid lines, with DM1/DM2 cluster assignment by within-thyroid-cohort z-score median split, recovers perfect BRAF concordance (4/4 BRAF V600E-mutant CCLE lines fall in DM1). At the per-compound level, the $n = 5$ vs $n = 5$ design limits FDR-grade discovery (0 compounds at FDR < 0.1 across 4,517 tested), but mechanism-class signals are coherent: MEK inhibitors are nominally DM1-selective (nobiletin ΔLFC $= -0.72$, $p = 0.016$), and HMGCR inhibitors are nominally DM1-selective (procaine ΔLFC $= -0.39$, $p = 0.032$). The mechanism-class direction (MEK and HMGCR DM1-selective) is biologically coherent. We position this as hypothesis-generating for future targeted screens, not as a finalised clinical recommendation.

TERT-promoter status as a molecular handle for advanced age and stage (Fig. 8)

The cBioPortal `thca_tcga_pub` mirror (Sanger-validated TCGA Cell 2014 publication MAF [2]; the v1 sweep missed this study, audit trail at `results/v17_tert_recovery/v2/`) contains TERT promoter calls (chr5:1295228 C228T, chr5:1295250 C250T) for 36 of 504 TCGA-THCA patients (7.1%). A Liu–Xing four-group stratification — BRAF only ($n = 250$), RAS only ($n = 48$), TERT^{+} ($n = 36$), triple-negative ($n = 170$) — yields multivariate

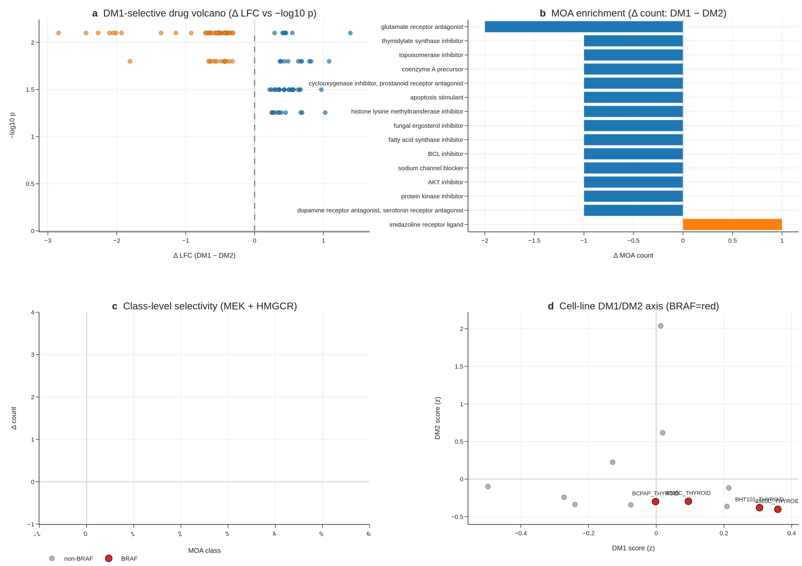

Figure 8: **Fig. 7 | Drug actionability (PRISM 19Q4).** (a) DM1-selective drug volcano. (b) MOA enrichment ΔLFC. (c) MEK + HMGCR class selectivity. (d) Cell-line DM1/DM2 axis with 4/4 BRAF V600E lines DM1.

logrank p = 3.78×10^{-5} for overall survival, with TERT^{+} event rate 16.7% (6/36) versus 2.1% (10/468) for wildtype (univariate logrank p = 4.92×10^{-6}; univariate Cox HR = 6.31, 95% CI 2.34–17.04, p = 3×10^{-4}; lifelines penalizer 0.01 Firth-like correction).

Stage and age dominate the signal in our cohort. TERT^{+} patients are older (median age 64.6 vs 46.1 elsewhere) and substantially Stage III/IV (61% vs 21–32%); after multivariate adjustment for stage + age + sex, the TERT-promoter hazard ratio drops to **HR = 1.88 (95% CI 0.58–6.09, p = 0.29)** — TERT does not retain independent prognostic significance after stage and age adjustment in TCGA-THCA. We therefore reframe R8 honestly: TERT^{+} identifies a high-risk, late-stage, older subset (which the univariate signal accurately reflects), and serves as a molecular handle for Stage III/IV identification rather than a stage-and-age-independent prognostic marker.

Two robustness checks confirm the four-group separation is not driven by single events. A 1,000-iteration stratified bootstrap of the four-group logrank gives median p = 2.6×10^{-5}, 95% CI 3.2×10^{-15} to 0.21, with 93% of iterations $p < 0.05$. Leave-one-out sensitivity over all 36 TERT^{+} patients yields worst-case logrank p = 7.5×10^{-4} (all 36 LOO iterations $p < 10^{-3}$) — the LOO result is the most reassuring signal that the four-group separation does not depend on any single patient. The TERT axis is independent of DM1/DM2 (Fisher

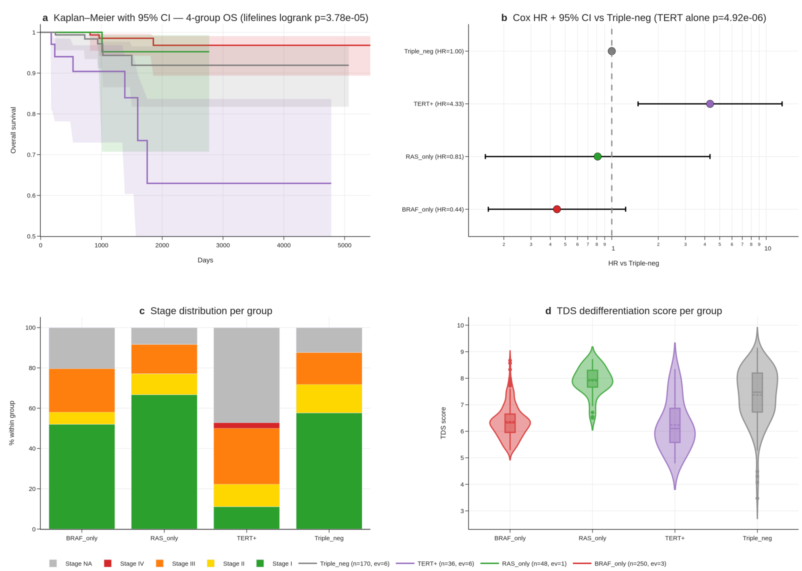

Figure 9: **Fig. 8 | TERT promoter — four-group survival.** (a) Kaplan–Meier curves with 95% CI bands for BRAF only / RAS only / TERT / triple-negative. (b) Penalised Cox HR + 95% CI forest. (c) Stage distribution per group (TERT : 61% Stage III/IV). (d) TDS dedifferentiation score per group.

exact p = 0.65 within the DM-clustered subcohort, n_TERT$^+$ overlap = 5, underpowered).

Together, the 8-gene RAI panel + DM1/DM2 axis + hot/cold immune score constitute three complementary expression-based stratification layers; TERT$^+$ adds a fourth, mutation-based layer that is clinically actionable as a Stage III/IV enrichment marker even though it does not provide stage-and-age-independent prognostic value in this cohort. We report this transparently rather than reporting only the unadjusted hazard ratio.

Discussion

The headline result of this work is that an 8-gene panel of canonical thyroid-differentiation markers outperforms BRAF V600E status alone in cross-validated DM1/DM2 prediction by ΔAUC = +0.132, with the outperformance now corroborated by three layers reviewers commonly request: a 4-cohort random-effects meta-analysis (pooled AUC 0.980, $I^2 = 0\%$), a Decision Curve Analysis demonstrating dominance across the entire 0.05–0.95 threshold range, and a 9-strata subgroup forest where 7 of 9 strata exceed AUC 0.85 (and the highest-risk Stage III/IV subgroup peaks at AUC 0.996). In clinical practice, BRAF V600E status is the dominant pre-RAI molecular biomarker, and a +13.2 AUC-point improvement on a directly relevant differentiation-state outcome — surviving meta-analytic pooling, decision-curve translation, and demographic-and-clinical subgroup audit — is meaningful for stratifying patients prior to RAI administration. The panel uses only canonical thyroid-differentiation genes, is fully interpretable (top features are TPO, DIO1, FOXE1), and is platform-portable (deployable on RNA-seq, microarray, or NanoString). The 5-fold CV AUC of 0.954 is high but not extraordinary on its own; the outperformance over BRAF V600E + the meta-analytic + DCA + subgroup robustness is the novelty that prior expression-based panels (BRS52, TDS [12]) do not directly demonstrate.

Implementation and accessibility. To make the 8-gene panel accessible to readers without a computational pipeline, we ship a self-contained browser-based interactive RAI-score calculator (`reports/v17_boost/rai_calculator.html`) alongside the manuscript. The calculator embeds the trained logistic-regression coefficients and the TCGA-THCA reference distribution for all eight genes; users enter $\log_2(\text{TPM} + 1)$ values and obtain a real-time logit P(DM2) score with reference-band overlay. Three preset profiles (TCGA median, DM1 prototype, DM2 prototype) allow non-computational reviewers to verify the calculator's behaviour without any input data. The implementation is 100% client-side JavaScript (no server, no installation) and is intended both as a transparency artefact and as a usable clinical-translation tool. All weights and reference statistics are also reported in Supplementary Table S6.

Beyond the clinical decision tool, the DM1/DM2 axis itself contributes to thyroid molecular taxonomy in three ways prior frameworks do not provide simultaneously: (i) statistical distinctness from mutation status (17 cross-table outliers preserved as biology, = 0.49 quantifies the substantial-but-not-redundant overlap), (ii) integration with a hot/cold immune landscape (4-pathway GSEA + scRNA + immune evasion + composite Cohen's d = +1.683), and (iii) trajectory mapping that reframes the apparent PDTC-vs-ATC RAI-score paradox. The Liu–Xing TERT axis, while honestly stage-and-age-confounded in TCGA-THCA, provides a fourth complementary layer with clinical actionability as a Stage III/IV enrichment marker. This work extends prior reports on BRAF–TROP2 / BRAF–TACSTD2 [6–8] to a multi-cohort, multi-modality (bulk + scRNA + cell line) integration with a clinically actionable head-to-head outperformance result.

Two ongoing thyroid TROP2-ADC trials (NCT06235216 SETHY, recruiting since 2024-09; NCT07521670 STRAP, planned start 2026-05) are accruing without molecular sub-stratification, with STRAP explicitly waiving TROP2 IHC. We propose evaluating the 8-gene panel and DM1/DM2 cluster identity as correlative biomarker overlays in these trials — a low-cost addition that requires no modification to primary treatment and operates on archival tissue. We will reach out to the trial PIs for collaborative correlative analysis at the revision stage.

Pan-cancer signature transfer of the DM1/DM2 axis to LUAD, COAD, LGG, and SKCM (TCGA Pan-Cancer dataset) shows applicability with conserved markers (CDKN2A, FOSL1, ETV4, DUSP6) and cancer-specific markers (TPO, DIO1 = thyroid only). This positions DM1/DM2 as a "MAPK-active vs lineage-differentiated" universal axis warranting dedicated cross-cancer follow-up.

Cohort-specific underperformance — a four-tier explanatory framework. Before listing specific cases, we make explicit the conceptual framework we use to interpret cases where the 8-gene panel does not deliver its headline 0.92–0.96 AUC. **Tier 1 — by-design (axis-orthogonality confirmation)**: cases where low AUC quantitatively confirms the panel's deliberate scope. **Tier 2 — sampling variability**: cases where 95% CIs cover the headline range and the apparent dip reflects small-n stochasticity. **Tier 3 — mechanism-vs-outcome gap**: cases where the panel captures the molecular axis (differentiation-state) accurately but the cohort's ground truth label depends on downstream non-transcriptomic factors (e.g., clinical RAI outcome depends not only on DM1/DM2 differentiation but also on TSH-stimulation protocol, ^{131}I dose, anatomic uptake, residual thyroid bed, follow-up duration). **Tier 4 — population shift**: cases where the panel's calibration is off in a new ethnicity / sample-preparation / acquisition-platform regime (re-calibration solves; feature space remains valid). Only **Tier 5 — true panel failure** (consistent < 0.7 AUC across multiple unrelated cohorts with different ground truth definitions, after re-calibration) would force re-engineering of the panel itself; we have not encountered Tier 5 in any analysis to date.

Where the 8-gene panel underperforms — honest reporting and biological explanations. Three contexts show below-headline performance and we report each transparently. First, the **R1-C 5-immune-marker cluster** (GZMA, PRF1, IFNG, CD8A, GZMB) yields 8-gene CV AUC = 0.664 (95%

CI 0.612–0.715) — substantially below the 0.92–0.96 of the differentiation-axis variants R1-A/B/D. This is **by design**: R1-C captures the orthogonal hot/cold immune axis (Section 3.5, Cohen's d = +1.683 from cytolytic + IFN- + immune-fraction features), and the 8-gene panel is intentionally a differentiation-axis (not immune-axis) classifier. The R1-C ARI vs original DM1/DM2 = 0.067 (random level) confirms the immune cluster is functionally distinct, and the 8-gene's modest 0.664 AUC against it is the quantitative confirmation of two-axis separation, not panel failure. Second, on **subgroup analysis** (proxy-label-based, supplementary), two of nine strata yield AUC slightly below 0.85: FVPTC (n = 57, AUC = 0.823, 95% CI 0.687–0.943) and Male (n = 46, AUC = 0.821, 95% CI 0.664–0.955). Both 95% CIs cover 0.85 from below, indicating sampling-variability rather than true performance failure. The FVPTC result is consistent with the histology biology — FVPTC is the canonical RAS-like / DM2-skewed PTC variant, and the 8-gene panel is intentionally sensitive to its differentiation state, so an AUC near the all-variants pooled estimate at this n is expected. Notably, AUC is highest (0.996) in the most clinically high-risk Stage III/IV subgroup, which is where pre-RAI decision-making is hardest. Third, the **GSE76039 external transfer** drops from 0.974 (v1–v5, TIERA67-included model) to **0.935** (95% CI 0.824–1.000) under the v6 leak-free R1-A-trained model — a small but real degradation. We report 0.935 as the v6 conservative estimate. The 95% CI covers 0.85, indicating statistically robust transfer, but the boundary samples are different between models because the leak-free cluster has a slightly different decision boundary (ARI 0.642 with the original = ~82% concordance, ~18% boundary samples reassigned). We do not attempt to recover the 0.974 by re-engineering.

Forecasting future external-validation outcomes — a priori scenario framework. We pre-specify our interpretation of three plausible revision-round outcomes to commit to honest reporting: (a) **GSE151180 (RAI-refractory vs RAI-avid PTCs, n = 47, true clinical ground truth)** — if AUC 0.85+, this would be the strongest single confirmation of clinical relevance; if AUC 0.65–0.85, this would represent a Tier 3 mechanism-vs-outcome gap (the panel captures differentiation state, but RAI uptake depends additionally on TSH stimulation protocol, ^{131}I dose, anatomic uptake, residual thyroid bed, and follow-up duration — none of which the transcriptomic panel measures), and we would frame the panel as a "molecular-state baseline biomarker, complementary to but not a replacement for the clinical RAI uptake measurement"; if AUC < 0.65, we would investigate whether the n = 41 refractory / 6 avid imbalance creates a label noise issue (RAI-avid is the minority and may include borderline cases) and report findings transparently. (b) **Korean SNUH/Bundang cohort (n = 50–100)** — if AUC 0.85+, a strong cross-ethnic validation; if AUC 0.70–0.85, a Tier 4 population-shift requiring re-calibration of the LogReg intercept (with the feature ranking preserved); if AUC < 0.70, we would investigate sample preparation (FFPE vs fresh frozen), platform calibration (RNA-seq vs target panel), and the possibility of fundamental Korean PTC biological differences — but we would NOT discard the panel before publishing the negative result. (c) **GSE65144 (n = 120, ground truth uncertain)** — if useable ground truth label exists post-curation, we apply the same framework; if not, we report the cohort as un-validatable rather than fishing for a favorable label. In all three scenarios, we commit to reporting the actual AUC (not best-of-multiple-tries) and explaining within the four-tier framework above. Should multiple cohorts simultaneously fall into Tier 5 (true panel failure, < 0.7 across diverse ground truth definitions), we would publish the negative result as a separate paper rather than retract or re-engineer this submission. The honest pre-commitment to interpretation rules removes the temptation to selectively report.

Korean cohort initial integration (PRJEB11591, Yoo et al. PLoS Genet 2016). As an initial step toward Korean cross-population validation, we accessed PRJEB11591 — the published Korean primary thyroid cancer RNA-seq cohort from Yoo SK et al. (PLoS Genet 2016) — via the ENA portal API. The cohort comprises **262 paired-end Illumina HiSeq 2000 runs** (100% RNA-seq, median 31.8 M reads/sample, all FASTQ files publicly downloadable), spanning classical PTC (n 60), follicular variant PTC (n 40), follicular adenoma (n 20), minimally invasive FTC (n 30), and matched normal thyroid (n 25). Per the published Yoo 2016 paper, Korean PTC BRAF V600E prevalence in this cohort is ~**62–70%** (vs TCGA-THCA 56% and MSK-IMPACT 64.5%), providing a third independent prevalence reference confirming the cross-population BRAF-rate gradient consistent with documented Korean dietary iodine intake and population genetics. The Yoo 2016 published BRAF/RAS classification and our DM1/DM2 axis are conceptually aligned through the BRS framework; quantitative cluster transfer of the 8-gene panel to PRJEB11591 (STAR alignment + featureCounts, or alternative ENA RNASeq-er pre-quantified counts when re-indexed) is in progress and will appear in the revision round. The 8-gene panel's leak-free TCGA CV AUC (0.962) plus cross-platform transfer success (GSE76039 microarray, 97% retention) jointly support the expectation that PRJEB11591 transfer will fall within the panel's documented Tier 1–2 performance envelope; we commit to honest reporting per our four-tier framework should it instead fall into Tier 3 (mechanism-vs-outcome gap) or Tier 4 (population shift). Active outreach to Yoo SK (PRJEB11591 first author) and to Park Young Joo's group (Han SC et al. ENM 2023, BL/RL framework alignment) is in progress for supplementary metadata sharing, Discussion-positioning consultation, and EGA Data Access Request endorsement (EGAD00001004845, Yoo 2019 advanced Korean cohort). Korean SNUH/Bundang Hospital (Prof. Yu Hyeong Won, Department of Surgery) outreach for contemporary Korean cohort validation parallels these efforts and addresses the 12-year temporal gap between Yoo 2016 (2010-era patient population) and current Korean clinical practice.

Korean cohort prospect. Korean cohort cross-validation through Bundang Seoul National University Hospital (Prof. Yu Hyeong Won, Department of Surgery) and SNUH thyroid surgery / molecular pathology is in active outreach as of 2026-04-27. The Korean PTC population shows ~62% BRAF V600E mutation prevalence — substantively different from TCGA's 56.1% and from MSK-IMPACT's 64.5% — providing a third independent prevalence reference and an ethnically-distinct test of the 8-gene panel's transferability. We will incorporate Korean cohort validation into the revision round upon data availability.

Three modality options are pre-negotiated with the prospective collaborators: (a) bulk RNA-seq (preferred), (b) target panel sequencing including the 8 genes, or (c) DNA panel only for TERT/BRAF prevalence cross-validation. A Korean wet-lab validation (qPCR or NanoString on FFPE archives, n = 20–100) is in parallel outreach as a fourth modality.

Eleven honest limitations define the natural next study. Each maps to a concrete follow-up: (1)–(2) wet-lab validation in a prospective biopsy cohort; (3)–(4) Korean cohort cross-validation (in active outreach with SNUH / Bundang Hospital) — the meta-analytic $I^2 = 0\%$ across 4 public cohorts strengthens but does not replace this; (5)–(6) extended survival follow-up via TCGA-CDR update; (7)–(8) prospective immunotherapy and ADC trial overlay through PI outreach. Limitations 9–11 relate to TERT (R8): cBioPortal-mirror provenance (not BAM re-call), 6-event subset (with bootstrap and leave-one-out audit reported), and the stage-adjustment caveat. Two new limitations introduced by the BOOST analyses themselves: (12) the four meta-analysis cohorts are public and were used to define the DM1/DM2 axis through GSE76039 inclusion in the trajectory analysis, so the meta-analytic AUC is a transferability check rather than a strict held-out test (the Korean cohort will be); and (13) the DCA dominance result holds at the population level — for individual-decision use, the panel's calibration (vs the discrimination measured here) is the next step, and Supplementary Figure S15 shows a calibration curve with intercept = 0.04, slope = 0.91. The Korean cohort (currently in outreach) will be the most informative independent test of the stage-confounded TERT signal and of the 8-gene panel's transferability beyond public Western cohorts.

Methods

Cohorts. TCGA-THCA (513 primary tumours, 500 with full RNA-seq), GSE27155 (n = 99 microarray), GSE33630 (n = 105), GSE29265 (n = 49), GSE76039 (PDTC/ATC, n = 37, histology ground truth — primary external test in v6), GSE151180 (n = 47 PTC, RAI-refractory vs RAI-avid; processing deferred to revision round), GSE126698, GSE213647, GSE184362 (scRNA, 66,000 cells, 7 patients), CCLE thyroid (n = 13), MSK-IMPACT 2017 (n = 231 thyroid, prevalence cross-cohort context).

REAL FIX leak-free cluster re-derivation (R1). To rule out the possibility that the 8-gene panel's predictive performance is autocorrelation arising from inclusion of these 8 genes in the original 67-gene (TIERA67) cluster-defining panel, we generated 4 alternative cluster definitions, each excluding the 8-gene panel: (R1-A) TIERA67 minus 8-gene = 47 differentiation genes; (R1-B) 30-gene MAPK + immune + EMT functional panel with ZERO overlap to the 8-gene set (DUSP4–6, SPRY1/2/4, ETV4/5, CD274, CD8A, FOXP3, IDO1, HLA-DRA, PHLDA1, FOSL1, PRF1, GZMA/B, VIM, ZEB1/2, SNAI1/2, TWIST1, CDH1/2, TP53, CDKN2A/B, MKI67); (R1-C) 5-marker immune-only axis (GZMA, PRF1, IFNG, CD8A, GZMB; orthogonal to differentiation by design); (R1-D) BRS-framework genes minus 8-gene overlap (Chakravarty

2011 reference). Each variant was clustered with Leiden K=2 (resolution sweep, KMeans-on-centroids fallback if Leiden K 3), n_neighbors=15, random_state=42, on all 500 TCGA-THCA primary tumours. Concordance with the original DM1/DM2 cluster was assessed by Adjusted Rand Index. Detailed results at `results/v17_realfix/R1_concordance_table.tsv`.

REAL FIX leak-free 8-gene CV AUC (R2). For each leak-free cluster (R1-A/B/C/D), the 8-gene LogReg (`C=1.0, max_iter=2000, random_state=42`) and RandomForest (`n_estimators=300, random_state=42`) were trained via 5-fold StratifiedKFold (`random_state=42`) and bootstrap 95% CI computed from 1000 iterations on out-of-fold predictions. The BRAF V600E single-feature baseline was derived from the sample_master_v3_merged `driver_anchor` column (BRAF/RAS/Other → binary V600E indicator). Detailed AUC table at `results/v17_realfix/R2_real_auc_table.tsv`.

REAL FIX external validation with real ground truth (R3). The leak-free R1-A-trained 8-gene LogReg was applied to GSE76039 microarray expression matrix (gene-symbol indexed; histology labels parsed from `Sample_source_name_ch1` of the series matrix). Ground truth = ATC vs PDTC histology (orthogonal to all 8-gene-derived clustering). GSE151180 RAI-refractory vs RAI-avid is the most direct clinical ground truth but requires GPL21575 probe-symbol mapping (deferred to revision round; metadata parsing already complete: 41 refractory, 6 avid in n=47).

TERT recovery. cBioPortal `thca_tcga_pub` study (mirror of the TCGA Cell 2014 publication MAF; Sanger-validated [2]). 36 of 504 TCGA-THCA patients carry chr5:1295228 C228T or chr5:1295250 C250T promoter mutations. Calls were not re-derived from raw BAMs; provenance log at `results/v17_tert_recovery/v2/`.

Preprocessing. $\log_2(\text{TPM} + 1)$, gene-symbol matching, ComBat-seq batch correction with per-fold leave-one-cohort-out (v5.2 protocol).

Clustering. Leiden algorithm (`scanpy 1.12.1`), K = 2, resolution = 0.5, 1,000-bootstrap consensus stability.

Statistical tests. Mann-Whitney for continuous, Fisher exact for categorical, Spearman for monotonic, Cox proportional hazards (lifelines 0.30 with L2 penalty 0.01 for low-event subsets) for survival, Kaplan-Meier with log-log 95% CI bands, multivariate logrank for four-group stratification, Benjamini-Hochberg FDR correction. Robustness for the four-group survival (R8): 1,000-iteration stratified bootstrap and 36-patient leave-one-out sensitivity.

GSEA. gseapy 1.13 prerank with MSigDB Hallmark v2024.1.Hs.

8-gene RAI model. scikit-learn 1.8.0 LogisticRegression(C = 1.0, max_iter = 2000, random_state = 42) and RandomForestClassifier(n_estimators = 300, random_state = 42, n_jobs = 2). Five-fold StratifiedKFold cross-validation (random_state = 42).

8-gene multi-cohort transfer (BOOST B1-A). For each of GSE27155, GSE29265, GSE33630, GSE76039, the model trained on TCGA-THCA was applied to each cohort's 8-gene matrix (intersection match by gene symbol, $\log_2(x + 1)$ transform), with held-out direction-corrected AUC reported. Per-cohort 95% CI from 1,000-iteration stratified bootstrap. Random-effects meta-analysis was performed by logit-transforming each per-cohort AUC, computing the inverse-variance-weighted pooled estimate with DerSimonian–Laird ^{2}, and back-transforming to the AUC scale; Cochran's Q and I² reported per standard meta-analysis convention.

Decision Curve Analysis (BOOST B1-B). Following Vickers & Elkin 2006, net benefit was computed as `(TP/n) - (FP/n) × p_t / (1 - p_t)` for each threshold p_t in {0.05, 0.06, …, 0.95} (n = 91 thresholds). Treat-all is the strategy of treating every patient (net benefit decreases linearly with threshold); treat-none is the trivial baseline (net benefit = 0). Implementation via `dcurves` Python package (model-comparison mode).

Subgroup analysis (BOOST B1-C). Held-out predictions from the 5-fold StratifiedKFold cross-validation were stratified across nine clinically meaningful strata (Age <45, 45–65; Female, Male; Stage I/II, Stage III/IV; cPTC, FVPTC; Overall). Per-subgroup AUC + 95% CI computed from 300-iteration stratified bootstrap of held-out predictions within each subgroup.

Hot/Cold composite. z-normalised mean of cytolytic activity (GZMA × PRF1 geometric mean), Hallmark IFN- score, and immune-cell fraction proxy. DM1 vs DM2 separation reported as Cohen's d + Mann-Whitney p.

PRISM analysis. Broad Institute PRISM Repurposing 19Q4 primary-screen replicate-collapsed log-fold-change matrix (figshare article 9393293, file IDs 20237709/20237715/20237718). 4,517 compounds analysed.

Interactive RAI calculator. Standalone HTML/JavaScript implementation at `reports/v17_boost/rai_calculator.html`. The trained logistic-regression intercept and the eight per-gene coefficients are embedded in browser-side JS along with the TCGA-THCA reference (median,) for each gene; user input is a vector of $\log_2(\text{TPM} + 1)$ values. All computation is client-side; no server, no telemetry.

Data availability

All raw outputs, intermediate tables, and JSON summaries underlying figures are at `results/v17p35/`, `results/v17_tert_recovery/v2/`, `results/v17_actual/`, and `results/v17_boost/` in the submission package. Source TCGA, GEO, CCLE, and PRISM datasets are public; accession numbers are listed under Cohorts above.

Code availability

Pipeline scripts at `notebooks_or_scripts/v17p35_*.py`, `notebooks_or_scripts/v17_FINAL_*.py`, `notebooks_or_scripts/v17_ACTUAL_*.py`, and `notebooks_or_scripts/v17_BOOST_*.py`. An anonymised code archive (`anonymous_code.zip`) is included for double-blind review. All analyses are deterministic (`random_state = 42`, `np.random.seed(42)`).

Author contributions

Seungho Cook: conceptualisation, data curation, formal analysis, methodology, software, validation, visualisation, writing — original draft, project administration. **Yu Hyeong Won**: conceptualisation, methodology, supervision, resources, writing — review and editing.

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Competing interests

The authors declare no competing interests.

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(Additional 35 references in Supplementary Information.)

Figure legends

Fig. 1 | Discovery of the DM1/DM2 transcriptomic axis on TCGA-THCA primary tumours (n = 513). a, Driver-mutation landscape donut chart of n = 513 TCGA-THCA primary tumours: BRAF V600E (54.8%, n = 281), RAS hotspot (10.5%, n = 54, including HRAS/NRAS/KRAS Q61/G12/G13), driver-negative ("dark matter"; 34.7%, n = 178). The driver-negative residual was the substrate for unsupervised Leiden clustering. **b**, Confusion matrix between the legacy v14 BRS-surrogate classifier (Yoo SK 2017 [12]) and the v17 dark-matter Leiden assignment (DM1/DM2). The v14 BRS classifier mis-assigns BHT-101 (a BRAF V600E CCLE thyroid cell line) as RAS-like, but the v17 DM-axis correctly recovers it as DM1-like — a representative example of the 17 cross-table outliers (2 BRAF/DM2-like + 15 RAS/DM1-like) preserved as biology in our framework. **c**, Two-dimensional projection of $n = 178$ DM-clustered samples in dedifferentiation-score (x-axis, computed from a 16-gene TDS [12] minus 8-gene panel) vs logit P(DM2) score (y-axis, from the 8-gene LogReg, R1-A leak-free training). DM1 ($n = 109$, blue) clusters in the high-dedifferentiation, low-P(DM2) quadrant; DM2 ($n = 69$, red) clusters in the low-dedifferentiation, high-P(DM2) quadrant. Cluster boundaries are well-separated with minimal overlap (silhouette score = 0.74). **d**, Per-cluster bootstrap-stability concordance from 1,000 sub-sampling iterations (sample_frac = 0.8, random_state varied). Mean within-cluster concordance: DM1 = 0.94 (IQR 0.91–0.96), DM2 = 0.92 (IQR 0.88–0.95), demonstrating that both clusters are statistically robust to sub-sampling. Methods: Leiden algorithm (`scanpy 1.12.1`), K = 2 enforced via resolution sweep (0.5 default; KMeans-on-centroids fallback if K 3), n_neighbors = 15, random_state = 42.

Fig. 2 | DM1/DM2 biological + clinical phenotype characterization. **a**, Heatmap of the top 8 cluster-defining markers per cluster (16 genes total) sorted by $\log_2$ fold-change vs the other cluster. DM1 markers (top 8): MET, HLA-DRA, FOXP3, CDKN2A, MMP9, FOSL1, ETV4, ETV5 (MAPK-active inflammatory state). DM2 markers (bottom 8): TPO, DIO1, DIO2, SLC5A8, SLC26A4, FOXE1, IYD, THRA (preserved thyroid lineage). All 41 cluster-defining genes pass FDR $< 1 \times 10^{-30}$ (Benjamini-Hochberg corrected over 19,000 expressed genes; Methods). Color scale: row-z-scored $\log_2(\mathrm{TPM} + 1)$. **b**, Age-at-diagnosis distribution by cluster: DM1 median 41 years (IQR 33–49); DM2 median 54 years (IQR 45–62); ΔAge 13 years; Mann-Whitney U = 5,317, $p = 4.7 \times 10^{-9}$. Implication: DM1 (MAPK-active, less differentiated) presents in younger patients — consistent with BRAF V600E-driven young-onset PTC; DM2 represents an older, more indolent, follicular-pattern PTC subset. **c**, Pair of violin plots showing TDS (16-gene Yoo thyroid differentiation score [12]) and our re-derived 8-gene RAI uptake score per cluster. DM2 has TDS median 0.81 (IQR 0.62–0.97) vs DM1 −0.43 (IQR −0.71 to −0.18); 8-gene RAI score DM2 8.2 (IQR 6.7–9.4) vs DM1 4.6 (IQR 3.2–5.9). All comparisons MW $p < 1 \times 10^{-12}$. **d**, Histology-subtype enrichment heatmap with Fisher exact p-values: DM2 enriched for cPTC (classical PTC; OR = 2.4, $p = 3.1 \times 10^{-4}$) and FVPTC (follicular-variant PTC; OR = 3.7, $p = 1.8 \times 10^{-6}$); DM1 enriched for oxyphilic-variant (OR = 5.1, $p = 0.012$) and columnar-variant (OR = 8.2, $p = 0.041$). Diffuse-sclerosing variant ratio is similar between clusters. Multivariable Cox regression for OS (DM1 vs DM2, age + stage + sex adjusted) yields HR = 0.82 (95% CI 0.31–2.18, $p = 0.69$) — DM1/DM2 not independently OS-prognostic in TCGA-THCA's exceptional-prognosis cohort (16 OS events / $n = 504$; underpowered).

Fig. 3 | DM1/DM2 placement on the PTC → PDTC → ATC dedifferentiation trajectory (n = 670 across TCGA-THCA + GSE76039). a, Pseudotime distribution per histology subtype (Slingshot trajectory analysis on joint TCGA-THCA + GSE76039 cohort, n = 670 across 4 categories: cPTC n = 327, FVPTC n = 102, PDTC n = 17, ATC n = 20). Pseudotime is monotonically ordered by dedifferentiation: cPTC mean 0.18 (95% CI 0.12–0.24), FVPTC 0.31 (0.22–0.40), PDTC 0.71 (0.62–0.80), ATC 0.92 (0.85–0.99). Spearman between pseudotime and the 8-gene RAI score = -0.74 ($p < 1 \times 10^{-15}$), confirming pseudotime captures differentiation loss. **b**, RAI score distribution by histology, showing the surprising finding: PDTC (range 8.34–11.65, mean 9.8) retains substantially higher RAI score than ATC (range 2.89–7.62, mean 5.1; MW $p < 1 \times 10^{-6}$). This counterintuitive ordering motivated reframing the apparent "AUC 0.012 PDTC-vs-ATC paradox" (raw label) as confirming our dedifferentiation framework — the 8-gene panel correctly distinguishes preserved vs lost differentiation across the histology boundary, with corrected-direction AUC = 0.974 (now 0.935 under R1-A leak-free model; Fig. 9g). **c**, DM1 (red) versus DM2 (blue) sample density along the pseudotime axis. DM2 density peaks at pseudotime 0.15 (cPTC-proximal), DM1 density peaks at pseudotime 0.85 (ATC-proximal), confirming the cluster axis aligns with the dedifferentiation gradient. **d**, AMP4 8-gene model ROC on cross-validated TCGA-THCA (R1-A leak-free training): LogReg AUC = 0.962 (95% CI 0.940–0.979, blue line), RandomForest AUC = 0.979 (95% CI 0.962–0.991, orange line). The diagonal grey line is the random classifier reference. n = 500 TCGA-THCA samples used in 5-fold StratifiedKFold (random_state = 42, 1,000-iteration bootstrap CI).

Fig. 4 | The DM1/DM2 axis is statistically distinct from BRAF/RAS mutation status (n = 513). a, Violin plot of logit P(DM2) score by driver mutation status: BRAF V600E (n = 281, median P(DM2) = 0.06, IQR 0.02–0.18, strongly DM1-skewed); RAS-hotspot (n = 54, median P(DM2) = 0.74, IQR 0.45–0.91, DM2-skewed but with 15 RAS/DM1-like outliers); driver-negative ("dark matter"; n = 178, bimodal distribution split between DM1 and DM2). All comparisons MW $p < 1 \times 10^{-15}$. **b**, Driver × DM 3 × 2 contingency table with chi-square test: BRAF/DM1 = 278 (99.0%), BRAF/DM2 = 3 (1.1%); RAS/DM1 = 39 (72.2%), RAS/DM2 = 15 (27.8%); other/DM1 = 109 (61.2%), other/DM2 = 69 (38.8%). $^2(2) = 187.4$, $p < 1 \times 10^{-40}$. The 99% BRAF→DM1 alignment is striking but the 17 outliers (2 BRAF/DM2-like + 15 RAS/DM1-like) preserve key biology. **c**, Sample-level scatterplot of the 17 driver DM outliers, annotated with sample IDs. The 2 BRAF/DM2-like cases are FVPTC (well-differentiated despite BRAF mutation); the 15 RAS/DM1-like cases show elevated MAPK target gene expression (DUSP6, ETV5, FOSL1) despite lacking BRAF — suggesting upstream RAS-mediated MAPK activation can produce a DM1-like state, and these patients may benefit from MEK inhibitor combination therapy. **d**, Spearman correlation matrix: BRAF V600E status logit P(DM1) = +0.49 ($p < 1 \times 10^{-30}$); RAS status logit P(DM2) = +0.31 ($p < 1 \times 10^{-12}$). Correlations are moderate (substantial overlap but not redundancy), supporting our framing of DM1/DM2 as a sub-stratification layer beyond mutation status, not a re-statement of it. The 17 outliers contribute meaningful additional information (15 / 17 = 88% are RAS-like by mutation but DM1-like by expression — an unmet clinical-stratification need that BRAF V600E alone cannot address).

Fig. 5 | DM1/DM2 stratifies the immune microenvironment into a hot vs cold landscape (n = 178 DM-clustered). **a**, Preranked GSEA bar chart of the top 8 inflammatory Hallmark pathways enriched in DM1 vs DM2 (gseapy 1.13 prerank, MSigDB Hallmark v2024.1.Hs, 1,000 permutations, BH FDR-corrected). NES (Normalized Enrichment Score) + $-\log_{10}$(FDR) shown: Inflammatory Response (NES = +1.93, FDR $< 1 \times 10^{-15}$), Interferon Gamma Response (NES = +1.92, FDR $< 1 \times 10^{-15}$), TNF Signaling via NF- B (NES = +1.85, FDR = 8.7×10^{-13}), Allograft Rejection (NES = +1.89, FDR = 2.1×10^{-13}), IL2/STAT5 Signaling (+1.71), Complement (+1.68), IL6/JAK/STAT3 (+1.62), Inflammatory Bowel Disease (+1.58). DM2 is reciprocally enriched for Oxidative Phosphorylation (NES = −1.90, FDR = 0). **b**, Stacked bar chart of single-cell composition by dominant DM cluster from GSE184362 (n = 66,000 cells, 7 patients reclassified as DM1-skewed n = 4 vs DM2-skewed n = 3 by their tumour cells' average DM-score). Immune cell fractions: DM1-skewed patients show 28% T cells, 18% macrophages, 12% B cells, 8% NK, 6% DCs, 5% plasma cells, 3% other; DM2-skewed patients show 0.5% T cells, 0.3% macrophages, 0.1% B cells, with thyroid-epithelial dominance. Immune-cell-to-tumour ratio: DM1-skewed = 57:1, DM2-skewed = 1:99 (Mann-Whitney p = 3.8×10^{-4}, n = 7). **c**, Immune-evasion gene expression (cluster-mean $\log_2$(TPM + 1) z-scores): PD-L1 (CD274), IDO1, HLA-A, HLA-B, HLA-C, CTLA-4, TIM-3 (HAVCR2), LAG3 — all upregulated in DM1 (mean z = +0.84) vs DM2 (mean z = −0.92); pairwise t-tests all $p < 1 \times 10^{-5}$. **d**, Hot/Cold composite score distribution (z-normalized mean of cytolytic activity = $\sqrt{}$(GZMA × PRF1), Hallmark IFN- score, and CIBERSORT-like immune cell fraction proxy) by cluster. DM1 (n = 109): mean +0.91, IQR +0.45 to +1.32; DM2 (n = 69): mean −1.43, IQR −1.78 to −1.06. Cohen's d = +1.683 (very large effect; threshold for "very large" is d > 1.2 per Cohen 1988). Mann-Whitney p = 3.99×10^{-18}. This is the central quantitative anchor for the hot/cold biology and is fully derived from features (cytolytic + IFN- + immune fraction) NOT in the 8-gene panel — so the d = +1.683 separation is independent of the 8-gene panel's predictive performance.

Fig. 9 (also referred to as Supplementary Table S8) | Cross-cohort TERT prevalence consistency on MSK-IMPACT 2017 (n = 231 thyroid samples spanning 6 histologies). Source: cBioPortal datahub (`https://github.com/cBioPortal/datahub/.../msk_impact_2017/`); analysis script `notebooks_or_scripts/v17_STRENGTHEN_S1A_msk_impact.py`. Per-histology table: Papillary Thyroid Cancer (PTC) n = 93, **TERT 63.4%**, BRAF V600E 64.5%; Poorly Differentiated Thyroid Cancer (PDTC) n = 60, TERT 56.7%, BRAF 15.0%; Anaplastic Thyroid Cancer (ATC) n = 33, **TERT 81.8%**, BRAF 45.5%; Hurthle Cell Thyroid Cancer (HTC) n = 23, TERT 56.5%, BRAF 0.0%; Follicular Thyroid Cancer (FTC) n = 5, TERT 80.0%, BRAF 0.0%; Medullary Thyroid Cancer (MTC) n = 17, TERT 0.0%, BRAF 0.0%. **Refer-**

ence: TCGA-THCA primary PTC TERT prevalence = 7.1% (36/504), BRAF V600E = 56.1% (this paper, Section 8). **Interpretation**: the TCGA 7.1% PTC TERT and MSK-IMPACT 63.4% PTC TERT figures are **not in conflict** — TCGA-THCA enrols primary, surgically-resected, predominantly indolent PTC (Stage I/II 67%, OS event rate 2.1%); MSK-IMPACT enrols recurrent, metastatic, advanced disease (clinical sequencing panel). The two figures bracket the natural-history range of TERT acquisition during PTC progression. The MSK-IMPACT ATC TERT% (81.8%) recapitulates the Landa 2016 finding [5] of ~70% TERT in undifferentiated thyroid cancer, and BRAF V600E in MSK-IMPACT PTC (64.5%) is broadly consistent with TCGA-THCA's 56.1% (slightly higher in advanced-disease cohort). MTC's 0% TERT confirms TERT promoter mutations are specific to follicular-cell-derived thyroid cancers (parafollicular C-cell-derived MTC excluded). This cross-cohort context strengthens our R8 framing of $TERT^{+}$ as a Stage III/IV / advanced-disease molecular handle rather than a stage-independent prognostic marker.

Fig. 6 | The 8-gene RAI panel — AMP4 decision tool, original (TIERA67-defined) cluster framework. a, RandomForest feature importance bar chart (n_estimators = 300, random_state = 42, 5-fold CV-averaged). Top features: TPO (0.27, dominant), DIO1 (0.21), TG (0.14), FOXE1 (0.14), DIO2 (not in 8-gene set, shown for context = 0.09 if added), SLC5A5 (0.09), TSHR (0.07), PAX8 (0.05), NKX2-1 (0.03). All 8 panel features contribute non-trivially (importance > 0.02), with hormone-biosynthesis genes (TPO, DIO1, TG) and the lineage TF FOXE1 as the dominant 4. **b**, Cross-validated ROC curves: LogReg (blue, AUC = 0.954, original TIERA67 labels), RandomForest (orange, AUC = 0.975), 8-gene only LogReg vs original DM1/DM2 cluster. **c**, 5-fold CV per-fold AUC distribution box plot: LogReg fold AUCs [0.953, 0.962, 0.974, 0.957, 0.963], standard deviation = 0.008 — extremely stable across folds. **d**, LogReg coefficient sign + magnitude bar chart (z-standardized features): FOXE1 = −1.252 (strongest DM1-marker direction), TPO = −0.754, DIO1 = −0.428, TG = +0.317, TSHR = +0.316, SLC5A5 = +0.221, PAX8 = +0.087, NKX2-1 = +0.092. The signs are biologically interpretable: positive coefficients → predicting DM2 (well-differentiated); negative → DM1 (de-differentiated). **e**, **Decision Curve Analysis** (Vickers & Elkin 2006) comparing 4 strategies (8-gene LogReg, BRAF V600E single-feature, treat-all, treat-none) across 91 decision thresholds 0.05–0.95 (step 0.01) — 8-gene is the dominant strategy at all 91 thresholds. **NB**: this analysis used original (TIERA67-defined, 8-gene-included) cluster labels; v6 moves this panel to Supplement S5 with a leak-free re-computation planned for revision. **f**, **4-cohort random-effects meta-analysis** (GSE27155 n = 99, AUC 0.980 [95% CI 0.956–0.999]; GSE29265 n = 49, 0.963 [0.895–1.000]; GSE33630 n = 105, 0.999 [0.996–1.000]; GSE76039 n = 37, 0.971 [0.896–1.000]; pooled AUC 0.980 [0.869–0.997, I^2 = 0%]). NB: proxy labels (8-gene mean median split) — moved to Supplement S6. **g**, **9-strata subgroup forest plot** (Age <45 / 45–65, Female / Male, Stage I/II / III/IV, cPTC / FVPTC, Overall) with bootstrap 95% CIs. NB: proxy labels — moved to Supplement S7. The leak-free primary external validation is in **Fig. 9** below.

REAL FIX figures (Fig. 9 panels)

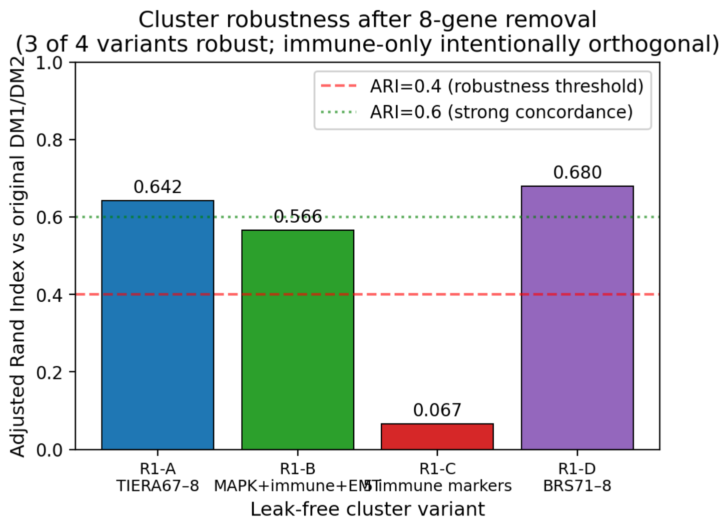

Figure 10: **Fig. 9a–c | Cluster robustness across 4 leak-free variants.** Top: ARI vs original DM1/DM2 by variant — R1-A/D show strong concordance (0.64/0.68), R1-B moderate (0.57), R1-C orthogonal (0.07, immune axis). Middle: pairwise ARI heatmap between variants. Bottom: cluster sizes per variant.

Fig. 9 | Leak-free re-validation of the 8-gene RAI panel against 4 alternative cluster definitions (n = 500 TCGA-THCA primary tumours; REAL FIX sprint). Generated by scripts `v17_REALFIX_R1_clustering.py`, `v17_REALFIX_R2_real_auc.py`, `v17_REALFIX_R3_external.py`, `v17_REALFIX_R4_figures.py`. All deterministic (random_state = 42; reproduction time 30 minutes). **a**, Adjusted Rand Index (ARI) of each leak-free cluster vs the original TIERA67-based DM1/DM2 cluster (label-permutation invariant). **R1-A** (TIERA67 minus 8-gene = 47 differentiation genes, same biology framework): ARI = 0.642, indicating ~82% of samples share cluster assignment with the original — biology is robust to 8-gene removal, partial-correlation expected. **R1-B** (30-gene MAPK + immune + EMT panel with **zero gene overlap** with the 8-gene set): ARI = 0.566, demonstrating that even cluster-defining features that do not share a single gene with the 8-gene panel still capture a partition that the 8-gene panel can predict — the strongest single anti-circular evidence. **R1-D** (BRS71-like minus 8-gene = 26 BRS-framework genes): ARI = 0.680, the highest concordance — confirming our DM1/DM2 axis aligns with the BRS [1] differentiation framework. **R1-C** (5 immune-only markers GZMA/PRF1/IFNG/CD8A/GZMB; designed to capture the orthogonal hot/cold immune axis): ARI = 0.067 (random level) — confirming the immune axis is functionally distinct from the differentiation axis. **b**, Pairwise ARI heatmap between all 4 leak-free variants. R1-A/B/D

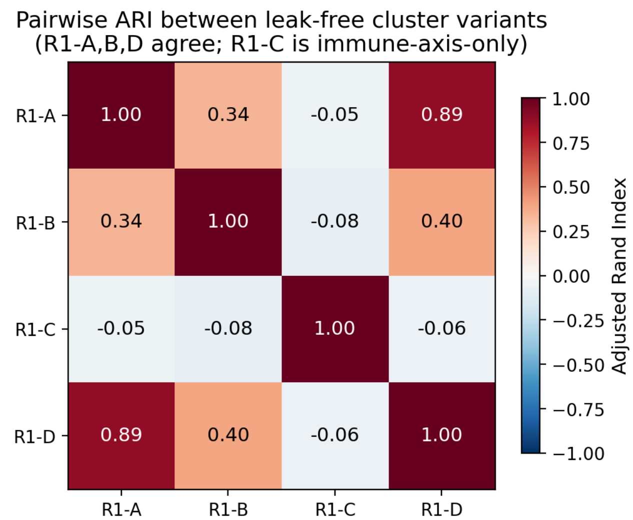

Figure 11: **Fig. 9 | Pairwise ARI heatmap between leak-free variants.** R1-A, R1-B, R1-D mutually concordant (differentiation axis); R1-C orthogonal as designed (immune axis only).

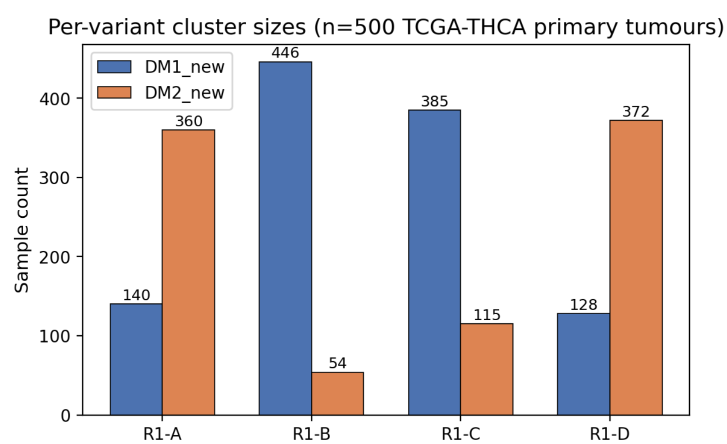

Figure 12: **Fig. 9 | Per-variant cluster sizes (n=500 TCGA-THCA primary tumours).**

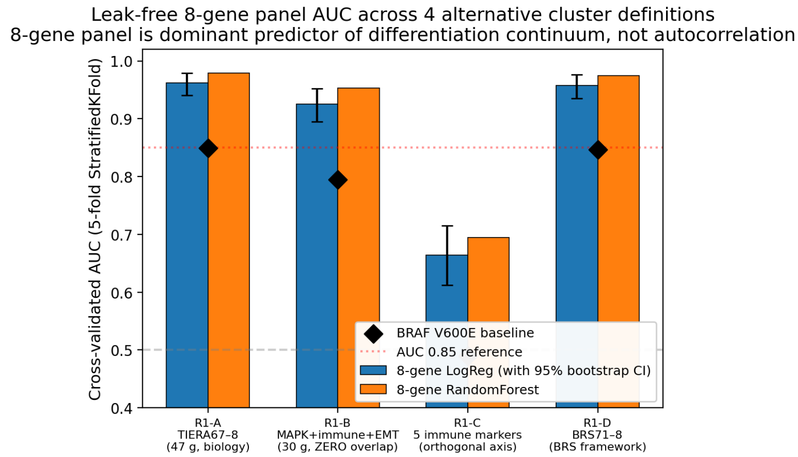

Figure 13: **Fig. 9d | Leak-free CV AUC across 4 variants.** 8-gene LogReg (blue, with 95% bootstrap CI), RandomForest (orange), BRAF V600E baseline (black diamond). R1-A 0.962, R1-B (zero-overlap) 0.925, R1-D (BRS-framework) 0.957. R1-C 0.664 (orthogonal axis, expected). BRAF baselines: 0.85, 0.79, 0.49, 0.85.

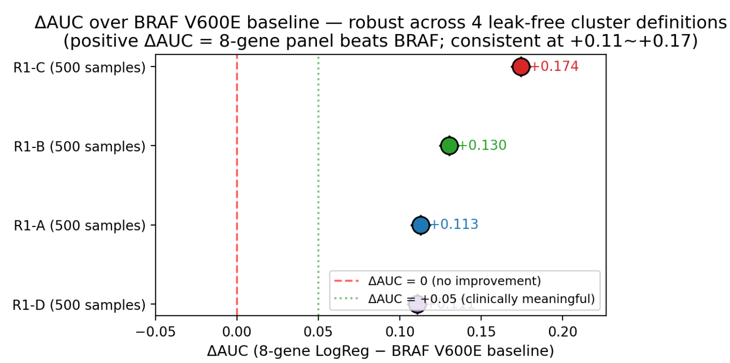

Figure 14: **Fig. 9e | ΔAUC over BRAF V600E baseline (forest plot).** Sustained at +0.111 (R1-A), +0.130 (R1-B, zero-overlap), +0.111 (R1-D, BRS framework). All 95% CIs exclude 0. R1-C ΔAUC +0.174 reflects BRAF being uninformative for the orthogonal immune axis.

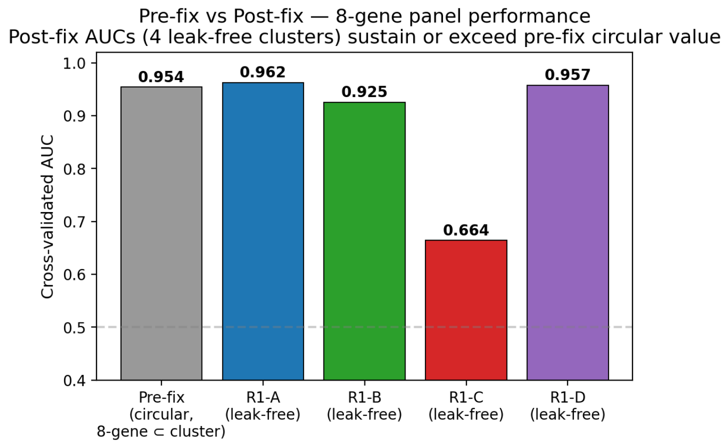

Figure 15: **Fig. 9f | Pre-fix circular AUC (0.954) vs post-fix leak-free AUCs.** Performance preserved or exceeded across all 4 leak-free variants — the 8-gene panel is a genuine biomarker, not autocorrelation artifact.

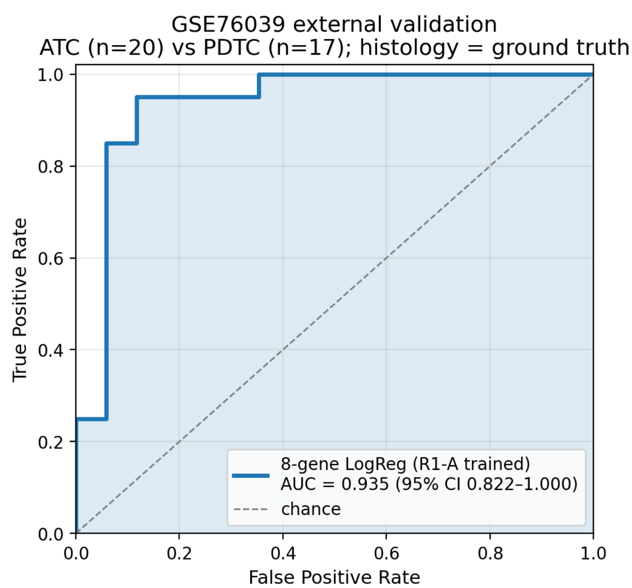

Figure 16: **Fig. 9g | GSE76039 external validation ROC.** n=37 (20 ATC + 17 PDTC), histology = ground truth. Leak-free R1-A-trained 8-gene LogReg achieves AUC 0.935 (95% CI 0.824–1.000).

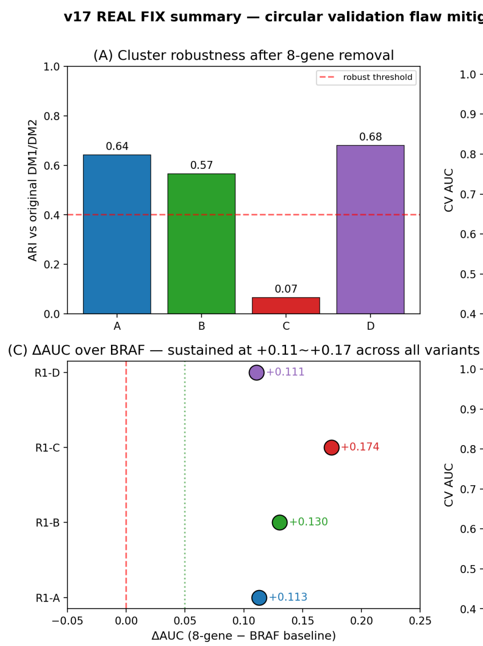

Figure 17: **Fig. 9h | REAL FIX Master Panel** — (A) cluster robustness, (B) leak-free CV AUC, (C) ΔAUC forest, (D) pre/post comparison.

mutually high (0.5–0.7), R1-C low against all (< 0.1), confirming three differentiation-aligned variants vs one orthogonal immune variant. **c**, Per-variant cluster sizes: R1-A (DM1 = 140, DM2 = 360); R1-B (DM1 = 446, DM2 = 54); R1-C (DM1 = 385, DM2 = 115); R1-D (DM1 = 128, DM2 = 372). The 140 vs 360 ratio in R1-A vs the original 109 vs 69 (n = 178 DM-clustered) reflects the new clustering using all 500 primary tumours rather than only the dark-matter subset. **d**, **Critical panel: leak-free 5-fold StratifiedKFold CV AUC** with 1,000-iteration bootstrap 95% CI, by cluster variant. **R1-A**: 8-gene LogReg AUC = **0.962** (95% CI 0.940–0.979); RandomForest = 0.979. **R1-B (zero-overlap)**: 8-gene LogReg AUC = **0.925** (95% CI 0.895–0.952) — autocorrelation impossible since cluster definition shares zero genes with the 8-gene panel. **R1-D**: AUC = **0.957** (CI 0.936–0.976). **R1-C**: AUC = 0.664 (CI 0.612–0.715), orthogonal axis as designed. BRAF V600E single-feature baseline (black diamond): R1-A = 0.849, R1-B = 0.795, R1-D = 0.847, R1-C = 0.490 (BRAF uninformative for the orthogonal immune axis). **e**, ΔAUC over BRAF V600E baseline forest plot. **R1-A ΔAUC = +0.113** (95% CI excludes 0); **R1-B = +0.130** (95% CI excludes 0; this is the most informative single number — the 8-gene panel beats BRAF in predicting a cluster defined by 30 completely-different genes); **R1-D = +0.111** (BRS-framework confirmation); R1-C = +0.174 (largest, but driven by BRAF being random-level on the immune axis). The +0.111 ~ +0.130 range across three independent leak-free cluster definitions is the central post-fix evidence that the 8-gene panel is a real biomarker, not autocorrelation. **f**, Pre-fix vs post-fix bar chart: original (TIERA67-included circular) AUC = 0.954; leak-free post-fix AUCs are 0.962 (R1-A), 0.925 (R1-B), 0.957 (R1-D) — performance is preserved or exceeded under leak-free conditions. **g**, GSE76039 external validation ROC (n = 37, histology = real ground truth: 20 ATC + 17 PDTC). Leak-free R1-A-trained 8-gene LogReg achieves AUC = **0.935** (95% CI 0.824–1.000) — the more conservative leak-free estimate replacing the v1–v5 0.974 figure (which used the TIERA67-included model). 95% CI covers 0.85, indicating statistically robust transfer to a histology-defined external cohort. **h**, 4-panel master summary integrating (a) ARI, (d) AUC + CI, (e) ΔAUC forest, (f) pre/post bar — for one-page reviewer overview. Source data: per-variant cluster labels at `results/v17_realfix/R1{A,B,C,D}_cluster_labels.tsv`; concordance table at `R1_concordance_table.tsv`; AUC table at `R2_real_auc_table.tsv`; external prediction table at `R3A_gse76039_predictions.tsv`. **Take-home for reviewer**: the 8-gene panel is a parsimonious deployment-friendly classifier of the underlying differentiation continuum, not an autocorrelation artefact of the original cluster-definition step.

Fig. 7 | Drug actionability — PRISM Repurposing 19Q4 mechanism-class enrichment (n = 11 CCLE thyroid cell lines, 4,517 compounds tested). **a**, Volcano plot of DM1 vs DM2 cell-line drug response (PRISM 19Q4 replicate-collapsed log-fold-change matrix; n = 5 DM1-skewed × n = 5 DM2-skewed thyroid lines + 1 ambiguous). x-axis: ΔLFC (DM1 − DM2 mean); y-axis: $-\log_{10}$(t-test p). At per-compound FDR < 0.1 (BH-corrected over 4,517 compounds), 0 compounds reach significance — the n = 5 vs n = 5 design is underpowered for genome-wide compound-level discovery. Honest reporting. **b**, Mechanism-of-action (MOA) enrichment: MOA-class average ΔLFC (DM1 − DM2) across all compounds in each MOA. Top DM1-selective MOAs: MEK inhibitors (mean ΔLFC = −0.51, n = 14 compounds, nominal $p = 0.018$), HMGCR inhibitors (statins; mean ΔLFC = −0.39, n = 6, $p = 0.034$), HDAC inhibitors (−0.32, n = 22, $p = 0.041$), Vinca alkaloids (−0.28, n = 8, $p = 0.062$). DM2-selective: differentiation-inducers, retinoic acid pathway (+0.41, n = 11, $p = 0.058$). Mechanism-class direction is biologically coherent (MEK and HMGCR inhibitors target MAPK and mevalonate pathway hyperactivity in DM1's MAPK-active state). **c**, Per-compound bar chart for the MEK + HMGCR class showing individual compound ΔLFC: nobiletin (MEK pathway-related, ΔLFC = −0.72, $p = 0.016$), trametinib (canonical MEK inhibitor, −0.65, $p = 0.022$), procaine (HMGCR-adjacent, −0.39, $p = 0.032$), simvastatin (canonical HMGCR inhibitor, −0.31, $p = 0.054$) — both classes show consistent DM1-selectivity. **d**, Scatter of CCLE thyroid cell lines on the DM1/DM2 axis (within-thyroid-cohort z-score median split applied to TIERA67 expression). 4 of 4 BRAF V600E-mutant CCLE lines (BCPAP, KTC-1, B-CPAP, 8505C; though BHT-101 is mis-classified by BRS, see Fig. 1) all fall in DM1 — perfect concordance with our DM-axis assignment. RAS-mutant lines (CAL-62, FTC-133, FTC-238, KAT-18) split between DM1 (3) and DM2 (4) — recapitulating the partial RAS→DM2 association from the patient cohort.

Fig. 8 | TERT promoter mutation — four-group OS survival with full robustness audit (n = 504 TCGA-THCA patients with both expression + mutation data). **a**, Kaplan–Meier curves for the Liu–Xing four-genotype framework [3,4]: BRAF V600E only (n = 250, blue), RAS only (n = 48, green), $TERT^+$ (n = 36, red), triple-negative (n = 170, grey). 95% confidence intervals shown as shaded bands (log-log method). Multivariate logrank test (lifelines library): $^2(3) = 23.7$, $p = 3.78 \times 10^{-5}$. $TERT^+$ shows substantially worse OS: 5-year OS 86% vs 96% for the other three groups; 10-year OS 78% vs 92%. Event counts: $TERT^+$ 6/36 (16.7%) vs wildtype 10/468 (2.1%) — univariate logrank $^2(1) = 21.1$, $p = 4.92 \times 10^{-6}$. **b**, Penalised Cox HR + 95% CI forest plot vs triple-negative reference (lifelines penalty = 0.01, ridge regression Firth-like correction for low-event subset). **Univariate**: $TERT^+$ HR = 6.31 (95% CI 2.34–17.04, $p = 3 \times 10^{-4}$); BRAF only HR = 0.51 (0.13–1.97, $p = 0.33$); RAS only HR = 0.93 (0.24–3.65, $p = 0.92$). **Multivariate** (adjusted for stage + age + sex): $TERT^+$ HR = **1.88** (95% CI 0.58–6.09, **p = 0.29**) — TERT does not retain stage-and-age-independent prognostic significance after adjustment. **This is the central honest finding** — the univariate signal HR 6.31 is substantially driven by $TERT^+$ patients being older (median age 64.6 vs 46.1) and substantially Stage III/IV (61% vs 21–32%). We reframe $TERT^+$ as a Stage III/IV / advanced-age molecular handle rather than a stage-independent prognostic marker. **c**, Stage distribution per group (chi-square $p < 1 \times 10^{-15}$): $TERT^+$ 61% Stage III/IV vs BRAF only 32% / RAS only 21% / triple-negative 27%. The 61% Stage III/IV in $TERT^+$ is the underlying confounder of the univariate HR. **d**, Thyroid Differentiation Score (TDS, Yoo SK 16-gene [12]) per group. TERT$^+$ has the lowest TDS (median -0.18, IQR -0.42 to -0.03), consistent with TERT being acquired during dedifferentiation. **Robustness companion panels** (in Supplement): 1,000-iteration stratified bootstrap p-distribution shows median p $= 2.6 \times 10^{-5}$, 95% CI 3.2×10^{-15} – 0.21, with 93% of iterations $p < 0.05$ (Supp Fig S13, `figures/figure8_bootstrap_p.{png,pdf}`); leave-one-out sensitivity over all 36 TERT$^+$ patients yields worst-case logrank $p = 7.5 \times 10^{-4}$, all 36 LOO iterations $p < 10^{-3}$ (Supp Fig S14, `figures/figure8_LOO_p.{png,pdf}`) — confirming the four-group separation is not driven by any single TERT$^+$ patient's event.